

A Source-Grounded Pipeline for Evaluating Legal Reasoning in Large Language Models

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Abstract—We study the problem of evaluating legal reasoning in large language model (LLM) answers when correctness depends on the path through a doctrine, not only on the final conclusion. We introduce a source-grounded evaluation pipeline that constructs a benchmark instance from a legal source case, elicits natural model answers, clusters those answers by inferred legal-reasoning signatures, and scores cluster representatives with a case-specific rubric. The pipeline contains five artifact-producing modules: Frank generates a source-grounded question, gold answer, and perturbations; Karthic derives a row-level rubric from the locked Frank packet; Dasha maps each natural answer to a normalized legal-reasoning signature and forms centroids; Judge applies the rubric to centroid representatives; and Zak records uncertainty or disagreement requiring escalation.

We validate the method on a real Illinois Statute-of-Frauds case, *Anglemire v. Policemen’s Benevolent Association of Chicago*. The live run uses ten response-model identifiers, two samples per model, and three Frank-generated question tracks: the original scenario, a legally invariant party-name edit, and a material signed-writing edit. Across 60 unlabeled natural answers, Dasha formed 23 track-aware reasoning centroids with complete member coverage (Wilson 95% CI for member/centroid coherence: 0.940–1.000). The invariant edit preserved the dominant reasoning family, whereas the material edit shifted the dominant answer family with a categorical distribution change under a permutation test (total variation distance 1.000, $p = 0.0005$). A two-model judge panel produced 230 final row-level scores; 18 rows required adjudication, and pre-adjudication agreement was moderate (weighted $\kappa = 0.630$, mean absolute error 0.465). Projected centroid scores produced model rankings, but bootstrap intervals overlapped at the top of the ranking. A separate 500-response deterministic regression suite verifies cluster bookkeeping and score projection over eight Statute-of-Frauds reasoning archetypes. The results support a bounded claim: the pipeline is reproducible, artifact-complete, perturbation-aware, and internally valid for the reported case study. Broader legal validity requires held-out cases and external expert review.

I. Introduction

Legal evaluation datasets usually expose models to a fixed prompt and then compare each answer with a reference answer or aggregate score. That design is often too coarse for legal reasoning. Two answers can reach the same bottom line through different doctrinal gates; two answers can cite the same doctrine but apply different exceptions; and two answers can discuss the same facts while resting on different legal theories. A legal benchmark therefore needs to measure reasoning paths as well as final outcomes.

We study source-grounded legal reasoning evaluation. The input is a citable legal source case, not a man-

ually authored hypothetical. The output is a complete evaluation bundle: a source-derived benchmark question, a gold answer, controlled factual perturbations, a case-specific rubric, natural model responses, reasoning-path clusters, centroid-level judge scores, projected member scores, model rankings, and escalation records. This formulation makes benchmark construction, response clustering, judging, and ranking part of one reproducible protocol rather than a sequence of informal manual decisions.

The reported implementation uses named LLM modules as artifact generators. Frank converts the source case into a benchmark packet. Karthic converts that packet into scored rubric rows. Response models answer the generated question without a system prompt or forced answer schema. Dasha abstracts each answer into a legal-reasoning signature and clusters signatures into centroids. Judge scores each centroid against Karthic’s rubric, and Zak records cases where uncertainty or disagreement prevents silent acceptance of the score. All module inputs, prompt hashes, outputs, and validation checks are serialized.

Statute of Frauds is the calibration domain for this paper because it has discrete doctrinal gates, boundary facts, and plausible alternative reasoning paths: marriage consideration, one-year performance, signed-writing compliance, suretyship, land, goods, executor promises, and equitable exceptions. The architecture itself is not Statute-of-Frauds-specific. The live path asks Frank, Karthic, and Dasha to infer doctrine, gates, rubric rows, and reasoning signatures from the source case and instruction stream. Statute of Frauds is therefore used as a litmus test for the general source-to-rubric-to-centroid method.

This paper makes a bounded methodological claim. We do not claim that the pipeline has established external legal validity across legal domains. We claim that the pipeline can be implemented as a reproducible artifact-generating method and that, on a real Statute-of-Frauds case study, it produces inspectable outputs at every stage: source-grounded Frank artifacts, a case-specific Karthic rubric, natural multi-model responses, perturbation-aware Dasha clusters, row-level judge scores, projected model rankings, and Zak escalation state. The appendices include direct excerpts from the live run bundle so that the reader can inspect the practical artifact quality rather than relying only on aggregate tables.

II. Contributions

The main contribution is a reproducible protocol for converting a legal source case into a clustered, rubric-scored evaluation of LLM legal reasoning. The protocol differs from static legal benchmarks in three ways.

Source-grounded benchmark synthesis. The benchmark instance is generated from a citable source case. The source-derived packet contains the question, gold answer, controlling legal gates, boundary perturbations, and provenance hashes used downstream.

Case-specific rubric induction. The scoring rubric is generated from the locked source packet rather than reused from a generic answer-quality checklist. Each row must correspond to a legally observable component of the case theory and include source support.

Reasoning-centroid evaluation. The pipeline clusters natural model answers by normalized legal-reasoning signatures before judging. A rubric-conditioned judge scores cluster representatives, and centroid scores are projected to all members to produce model-level rankings with row-level rationales.

These contributions separate three operations that are frequently conflated in LLM evaluation: constructing a legal benchmark, identifying families of legal reasoning in model outputs, and scoring those families under a source-grounded rubric.

A. Statute-of-Frauds Litmus Test

Statute of Frauds is the first litmus domain because it naturally contains discrete legal gates and boundary facts: marriage consideration, one-year performance, suretyship, land interests, goods/UCC writings, executor or administrator promises, writing substitutes, and equitable exceptions. That structure makes it a useful early test for whether an automated pipeline can generate legally meaningful variations and separate different reasoning paths.

In the present validation, the strongest evidence concerns pipeline mechanics and representative Statute-of-Frauds reasoning paths. The live model batch exercises the full source-to-ranking path on a marriage-consideration beneficiary-certificate case, but because the sampled models converged on one wife/certificate theory, it is not a sufficient test of Dasha’s divergence-detection claim. The 500-response regression suite checks the Dasha clustering container against eight Statute-of-Frauds archetypes: wife/certificate enforcement, association replacement-certificate control, one-year/no-writing bar, one-year-under-year enforceability, suretyship/main-purpose exception, land part-performance, goods/UCC quantity defect, and executor personal-promise without writing. This does not prove every possible Statute-of-Frauds case is solved, but it shows that the pipeline’s schemas, clustering machinery, and validation surfaces cover the major gate families needed for that domain.

B. Generalization Beyond Statute of Frauds

The architecture is doctrine-general because every live stage is driven by the source case and canonical instruction context. Frank is asked to infer the doctrine and legal gates; Karthic builds a rubric from the locked Frank packet; Dasha normalizes whatever doctrine, trigger, outcome, and exception structure is present in the responses; Judge scores against Karthic’s rows rather than a Statute-of-Frauds-specific checklist. To apply the method to another legal field, the expected change is not the pipeline architecture but the instruction context and calibration examples supplied to the agents. Broader legal-domain claims therefore require additional source packs and held-out cases, but the pipeline is already structured so those additions are data and instruction extensions rather than a new evaluation system.

III. Related Work

The pipeline builds on LLM-as-judge research, but adapts it to source-grounded legal reasoning rather than general helpfulness or preference evaluation. MT-Bench and Chatbot Arena showed that strong LLM judges can approximate human preferences at scale, while also surfacing biases such as position, verbosity, and self-enhancement effects [1]. G-Eval introduced a rubric-like evaluation procedure using chain-of-thought-style criteria and reported stronger human alignment for natural-language generation evaluation [2]. Prometheus and Prometheus 2 further support the idea that rubric-conditioned evaluator models can provide fine-grained judgments when given reference material and score criteria [3], [4].

The judge literature also makes clear why internal validation must test more than whether an LLM can emit a score. Position-bias studies show that pairwise or listwise judges may change preferences when candidate order changes [5]. Self-preference work shows that judge models may favor outputs from their own model family [6]. For legal evaluation specifically, recent benchmarks and legal judge proposals emphasize that legal reasoning requires decomposed factual and doctrinal units, not just general answer preference. LegalBench demonstrates the value of law-professional task design and explicit legal-reasoning categories across a large task suite [7]. LEXam uses law-exam questions with reference answers and reasoning guidance, and reports an ensemble LLM-as-judge setup with human expert validation [8]. LeMAJ decomposes legal answers into legal data points and evaluates them in a way intended to reflect legal expert review practices [9]. These findings motivate the row-level rubric design, centroid-level judging, model-output provenance, and explicit separation between internal validation and future expert-reviewed legal validity.

The perturbation lane is related to metamorphic testing, where correctness is tested through expected relations between multiple inputs and outputs rather than through

a single oracle [10]. In this pipeline, Frank’s invariant and material variations act as legal metamorphic relations: legally irrelevant edits should preserve the dominant answer path, while legally operative edits should change the dominant outcome or reasoning path when the doctrine requires it.

The pipeline differs in three ways. First, the benchmark is generated from a legal source case rather than from a static task prompt. Second, responses are clustered by legal reasoning path before judging, allowing many model samples to be compressed into interpretable centroids. Third, the judge scores rubric rows on centroids and then projects those scores to every response in the cluster. This makes model ranking cheaper and more interpretable while preserving row-level evidence for review.

IV. Methods

A. Problem Formulation

Let s denote the text of a legal source case and let $M = \{m_1, \dots, m_n\}$ be a roster of response models. The goal is to estimate how the models reason about a source-derived legal question while preserving the relation between source authority, benchmark construction, answer clusters, and scores. The pipeline constructs a sequence of artifacts:

$$s \rightarrow F \rightarrow R \rightarrow T \rightarrow Y \rightarrow C \rightarrow S \rightarrow A.$$

Here F is the Frank packet containing the generated question, gold answer, doctrinal gates, and perturbations; $R = \{r_1, \dots, r_k\}$ is the Karthic rubric; T is the set of executable question tracks; Y is the set of raw model answers; C is a partition of Y into reasoning clusters; S is the row-level judge score matrix over cluster representatives; and A is the set of escalation and audit artifacts.

For each model m , track t , and sample j , the response generator produces a natural answer $y_{m,t,j}$. Dasha maps each answer to a signature $\sigma(y)$ containing canonical doctrine, rule-trigger, outcome, exception-or-defense, and primary-reasoning identifiers. Clustering is a track-conditioned equivalence relation:

$$y_a \sim y_b \iff t_a = t_b \wedge \sigma(y_a) = \sigma(y_b).$$

Judge then scores one representative response from each cluster against every rubric row. If w_i is the weight of row r_i and $q_{c,i} \in [0, 4]$ is the final judge score for cluster c , the projected score for any response in cluster c is

$$\text{score}(y) = \sum_i w_i q_{c,i}.$$

Model rankings aggregate projected scores across that model’s samples and question tracks. This formulation makes the central validity condition explicit: score projection is meaningful only when the cluster relation groups legally equivalent reasoning paths.

B. Pipeline Overview

The reference implementation is a command-line research harness with an optional inspection interface. Runs are controlled by JSON configuration files that specify a source-case path, output directory, model identifiers, sample counts, temperatures, quality gates, clustering method, perturbation policy, and judge settings. The harness writes every stage artifact as JSON: Frank packet, Karthic rubric, model responses, Dasha clusters, judge scores, Zak packets, manifest, validation summaries, and paper tables. This design makes the research claim reproducible from recorded artifacts rather than from an interactive UI session.

For claim-supporting runs, the source-case path must refer to a real, citable court-case source and must have a metadata sidecar. The sidecar records case id, title, citation, jurisdiction, court, decision date, source type, provenance, calibration role, material facts, expected legal pressure points, and limitations. The live Statute-of-Frauds configuration uses the Anglemire court case PDF; compact text fixtures remain only for offline smoke tests and regression tests. When the source is a PDF, the harness extracts text for LLM input while hashing both the original PDF bytes and extracted text for provenance.

The pipeline has six functional stages. Frank ingests a source case and produces a doctrine-aware benchmark packet. Karthic generates a rubric from the locked Frank packet. The perturbation layer converts selected Frank variations into executable question tracks. Response generation collects natural model answers for each track. Dasha clusters generated model responses by legal reasoning signatures. Judge scores Dasha centroids against Karthic rows and projects those scores to all members of each cluster, while Zak produces escalation packets when uncertainty or disagreement warrants further review. The stages communicate only through serialized artifacts, which makes each boundary auditable and testable.

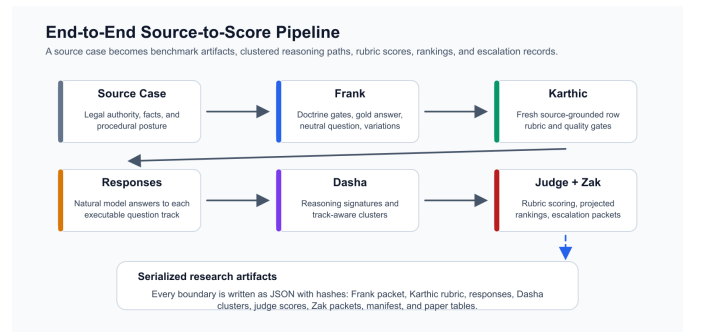


Fig. 1. End-to-end source-to-score pipeline. A legal source case is converted into auditable benchmark artifacts, natural model responses, Dasha clusters, rubric-conditioned judge scores, projected rankings, and Zak escalation records.

C. Model-Call and Provenance Layer

All live LLM calls use a small JSON-over-HTTP client rather than a framework agent runtime. The model-call layer has two functions: `generate_text` for ordinary completions and `generate_json` for strict JSON-producing agents. When an agent returns malformed JSON, `generate_json` attempts a repair pass and then re-parses the result. This keeps Frank, Karthic, Dasha, and Judge on explicit schemas instead of free-form prose.

The agent architecture is stateless and artifact-mediated. Frank, Karthic, Dasha, and Judge are not long-running autonomous processes, do not call tools, and do not share conversational memory. Each stage is one or more model calls whose complete inputs are the stage instruction stream, the relevant upstream JSON artifact, and the run configuration. The stage output is parsed into a versioned JSON artifact and becomes the only input passed downstream. This is deliberate: the experiment should be reproducible from source text, prompt hashes, model identifiers, configuration values, and JSON artifacts rather than from an opaque chat history.

The implementation distinguishes legal judgment from reproducibility plumbing. Frank, Karthic, Dasha, and Judge are LLM agents that make the semantic legal decisions from the source case, instruction context, and generated answer. The deterministic code path is limited to schema validation, prompt hashing, manifest construction, budget enforcement, provenance recording, and exact grouping over canonical fields emitted by Dasha. Response models are sampled with nonzero temperatures and repeated samples because variability in natural legal answers is part of the object being measured. The reported live run used gpt-5.2 for Frank and Karthic, claude-sonnet-4-6 for Dasha signature extraction and id canonicalization, nonzero response temperatures between 0.55 and 0.65 with two samples per model per track, and a low-temperature two-model claude-sonnet-4-6/ claude-haiku-4-5-20251001 judge panel with a claude-sonnet-4-6 adjudication pass for unstable rows. Runtime adapters are treated as execution plumbing; the research unit is the named model output and its recorded prompt provenance.

Every run records provenance at three levels. First, source text is hashed with a short SHA-256 content hash. Second, each live prompt bundle is hashed after including the instruction context and agent settings. Third, generated responses record model identifier, sample index, question id, response prompt style, timestamp, and text. The purpose is not merely bookkeeping: prompt hashes make it possible to distinguish a substantive methodological change from a repeated run under the same frozen instructions.

D. Instruction Context

The instruction context is not an implicit memory store or a hidden code path. It is an ordered text bundle inserted into the prompt of each pipeline agent. At runtime, the

TABLE I
Model roles in the reported live run.

Pipeline role	Model identifier	Function
Frank	gpt-5.2	Convert source case into doctrine gates, neutral question, gold answer, variations, and controller card.
Karthic	gpt-5.2	Convert the locked Frank packet into a source-grounded row-level rubric.
Dasha extractor and canonicalizer	claude-sonnet-4-6	Read each model response, emit a normalized legal-reasoning signature, and merge equivalent canonical ids before clustering.
Judge	claude-sonnet-4-6; claude-haiku-4-5-20251001	Score each Dasha representative against every Karthic rubric row; adjudicate unstable panel rows with claude-sonnet-4-6.
Response models	Ten model identifiers including GPT, Claude, Gemini, DeepSeek, and Llama families	Answer the original and perturbed Frank-generated questions before clustering.

harness reads a fixed list of versioned prompt components, prepends each excerpt with a stable component identifier, concatenates the excerpts up to a declared budget, and records a hash of the resulting bundle in the stage artifact. The resulting string is what this manuscript calls the instruction stream: an ordered prompt prefix that contains the general legal reasoning protocol plus the role-specific operating rules for the current agent. Thus, the legal and procedural constraints available to Frank, Karthic, and Dasha are reproducible text inputs rather than informal background knowledge.

The first component in every agent bundle is a cross-agent legal reasoning protocol. That protocol states that Statute of Frauds is the first calibration domain, not the global assumption; that source authority controls legal substance; that schema terms such as `doctrine_gates` and `exception_or_defense` refer to the controlling elements of whatever doctrine the source supports; and that uncertainty should be escalated rather than filled by forcing a familiar doctrine. After this shared protocol, each agent receives role-specific instructions. Frank receives source-intake, packet-building, and question-writing rules. Karthic receives rubric-construction and scoring-overlay rules. Dasha receives centroid-evaluation, judge-panel, citation-checking, and Zak-escalation rules. The appendix at the end of the manuscript describes the contents of these bundles in detail.

E. Frank: Source-to-Benchmark Generation

Frank receives the source case text and is instructed to infer the doctrine, legal gates, material facts, source limits,

neutral question, gold answer, and boundary variations. The gold answer uses a consistent structure: jurisdiction assumption, bottom-line outcome, controlling doctrine, transaction characterization, writing or element trigger, compliance or exception analysis, other defenses, and strongest counterargument. This structure is doctrine-aware: for non-Statute-of-Frauds cases, the heading content should map to the appropriate doctrinal elements rather than force an irrelevant writing analysis. The live Frank prompt loads the frozen instruction stream, including the cross-agent general legal reasoning protocol, and expressly treats Statute of Frauds as a calibration domain rather than a default doctrine.

Technically, Frank returns strict JSON with fields `doctrine_family`, `detected_doctrine_gates`, `source_extraction`, `neutral_question`, `gold_answer`, `variations`, and `controller_card`. The implementation then wraps that output in a versioned `research.frank.llm.v1` packet, attaches source and prompt hashes, and promotes `controller_card.primary_gate_id` as the downstream control field. A backward-compatible `statute_of_frauds` object is emitted only when the inferred doctrine is actually Statute of Frauds. The reason for this schema is that later stages should depend on explicit source-grounded doctrine gates rather than on hidden prompt assumptions.

F. Karthic: Dynamic Rubric Construction

Karthic builds a fresh rubric by converting the locked Frank packet into a modular scoring instrument. The input is not only the neutral question. Karthic receives the full Frank packet: source hash and excerpt, inferred doctrine family, detected doctrine gates, source-extraction facts, gold answer, variations, primary gate id, strongest counterargument, and controller-card status. It also receives the Karthic instruction stream, which specifies a fixed build procedure. The purpose of this stage is to turn Frank’s legal theory into observable scoring rows while preserving source support and avoiding double-counting.

The construction procedure has six steps. First, Karthic performs a packet audit. It checks that the current question, controlling doctrine, trigger test, trigger facts, variation lane, and selected variation are mutually consistent. Fields are treated as fixed, fixed but jurisdiction-sensitive, or needing confirmation. Second, it locks the doctrinal path by writing the controlling doctrine, required gate order, benchmark posture, and strongest expected counterargument. If those cannot be stated from the packet, the rubric should not be locked. Third, it instantiates the fixed parent module shell: metadata tags, structural gatekeeping, primary doctrine gates, fallback doctrines and defenses, and cross-cutting answer discipline. Metadata is not scored.

Fourth, Karthic drafts coarse scored rows inside those modules. Rows are intended to cover the legally meaningful path rather than generic answer quality. The required

coverage includes doctrine or gate, rule or element, facts, compliance, exceptions or defenses, counterarguments, conclusion, variation sensitivity, and source support when applicable. The full instruction spec describes each row as a row card with a code, module, weight, description, non-applicability guidance, golden target summary, observable features, allowed omissions, contradiction flags, comparison guidance, 0–4 scoring anchors, failure labels, and row status. The reference implementation stores the scoring core of that card: `id`, `category`, `weight`, `criterion`, and `source_support`. This distinction matters: the method is designed around full row cards, while the present JSON artifact materializes the fields needed for automated scoring and validation.

Fifth, Karthic decomposes, prunes, and audits overlap. It splits a broad row only when the split improves legal discrimination, such as when one row merges a trigger analysis with an exception analysis or collapses two distinct doctrinal gates. It merges or removes rows that are stylistic, duplicative, too correlated, or better represented as failure-label modifiers. For close rows, the instruction spec requires a distinctness rationale so that one underlying legal error is not counted twice.

Sixth, Karthic assigns weights and variation handling. The instruction spec starts from module-level defaults of 28 percent for structural gatekeeping, 40 percent for primary doctrine gates, 19 percent for fallback doctrines and defenses, and 13 percent for cross-cutting answer discipline. The controlling doctrine row should usually be the heaviest row, secondary gates should not outweigh the controller, fallback rows should remain secondary unless the source makes them central, and Module 4 should stay light. Variation notes are tied only to the selected Frank variation: answer-invariant Lane A variations preserve doctrine, outcome, gate order, and weights, while Lane B missing-fact variations may add or strengthen rows about bounded uncertainty and conditional branching.

The rubric artifact is versioned as `research.karthic.llm.v1`. The live implementation asks Karthic, using gpt-5.2 at temperature 0 in the reported run, for strict JSON with a top-level rows array. Each row must contain an `id`, `category`, `weight`, `criterion`, and `source-support` field. The parser accepts common alternate LLM shapes such as `rubric.rows`, `base_rubric.rows`, and module-level row arrays, then normalizes semantic category aliases: for example, “doctrine/gate”, “doctrine_gate”, and “gate” are all mapped to `doctrine`; “exceptions/defenses” maps to `exceptions`; and “source” maps to `source_support`.

After parsing, the harness applies automated quality gates. It checks minimum row count, required category coverage, missing row ids or criteria, and duplicate-like criteria using token-level Jaccard similarity. If the first rubric fails minimum row count or required category checks, the harness runs a strict repair prompt requiring at least the configured number of rows, all required

TABLE II
Karthic rubric-construction procedure.

Step	Operation	Output used downstream
Packet audit	Check Frank question, doctrine, trigger facts, variation lane, and controller-card fields for consistency.	Fixed fields and review flags.
Path lock	State controlling doctrine, gate order, benchmark posture, and strongest counterargument.	Rubric theory of the case.
Module shell	Instantiate metadata, structural gatekeeping, primary doctrine, fallback, and answer-discipline modules.	Stable row organization.
Row drafting	Create source-grounded rows that observe legal behavior rather than prose style.	Scored row candidates.
Decompose and audit	Split, merge, prune, and document overlap to prevent double-counting.	Non-duplicative row set.
Weight and patch	Assign legally important weights and patch only the selected variation track.	Final rubric rows and scoring policy.

categories, doctrine-specific source-grounded criteria, row ids, weights, and source-support fields. If weights do not sum to one, they are renormalized across rows. This repair step is engineering calibration rather than evidence of legal validity, but the final artifact still records the prompt hash and schema so the frozen rubric can be reproduced.

G. Response Generation

The pipeline evaluates a roster of concrete model identifiers. The reference implementation can route calls through different execution backends, but the research unit is the model output, not the backend used to obtain it. Every model receives the same Frank-generated question. By default, response models receive no system prompt, no source excerpt outside the question, and no required heading structure. The user message is the Frank question itself. This was chosen because the scientific object is the model’s natural legal response, not its ability to fill a researcher-imposed memo template. Each response records the model identifier, sample index, timestamp, question id, response prompt style, and prompt provenance.

The reported live natural-response batch used `response_prompt_style = natural`: each model received only the Frank question and answered in its own format. Temperature is controlled in the run config and offset slightly across repeated samples from the same model to induce natural answer variation without changing the question or rubric. No response is given a target reasoning label before generation.

H. Question Perturbation Tracks

Frank’s variations are not treated as decorative examples. The implementation converts the original question and selected Frank variations into executable question tracks. Each track has a stable `track_id`, `question_id`, edited question text, perturbation type, changed fact, and expected behavior. The original question is the base track. Invariant perturbations change legally irrelevant surface facts, such as party or company names; the expected behavior is that the dominant answer path remains stable. Material perturbations change legally operative facts, such as duration, writing, price, beneficiary designation, possession, or default posture; the expected behavior is that the dominant answer path or reasoning signature changes.

This design follows the logic of minimal-pair and counterfactual robustness testing: edits should be as small as possible, and the expected effect of the edit should be stated before model responses are generated. In the live path, the same response roster can answer each track using the natural question-only prompt. Dasha then clusters responses separately by track so an answer to a variant cannot be hidden inside the original-question cluster. The perturbation report compares each variant’s dominant Dasha signal to the original track. Invariant tracks pass when the dominant outcome and reasoning path are preserved; material tracks pass when the dominant outcome or reasoning path differs. This gives the pipeline a built-in test for over-sensitivity to irrelevant facts and under-sensitivity to legally controlling facts.

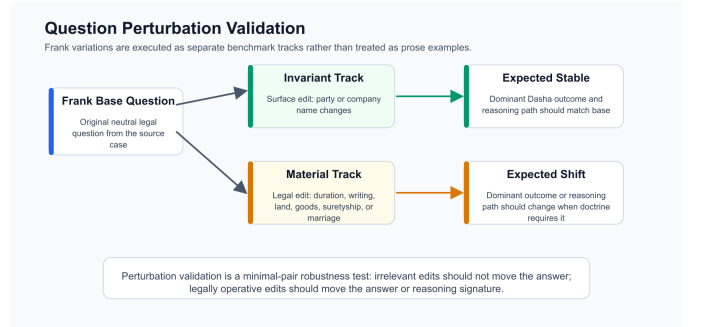


Fig. 2. Question perturbation validation. Frank’s base question and selected variations become separate executable tracks. Invariant edits are expected to preserve the dominant Dasha signal; material legal edits are expected to change the dominant outcome or reasoning path when the doctrine requires it.

I. Dasha: Reasoning-Signature Clustering

Dasha has two distinct responsibilities in the live path: semantic extraction and grouping. First, for each generated response, Dasha receives three inputs: the Dasha instruction stream, a truncated copy of the locked Frank packet, and the raw response text. It is prompted to return strict JSON with the keys `doctrine`, `issue`, `rule_trigger`, `outcome`,

exception_or_defense, primary_reasoning_path, reasoning_path, secondary_paths, conclusion, and key_distinguishing_facts. The primary path is the controlling path on which the answer ultimately rests. The secondary path list records material gates, exceptions, equitable theories, or counterarguments that the response accepts, rejects, treats as uncertain, or otherwise discusses before reaching the final conclusion. This extraction step is the semantic abstraction layer: it asks the model to state what legal path the response appears to use, without telling the response model which path to use and without giving Dasha a preassigned label.

Second, the implementation normalizes the extracted signature before clustering. The sanitizer removes unexpected non-Latin script fragments and collapses whitespace. In the live path, Dasha emits doctrine-general canonical identifiers: `doctrine_id`, `rule_trigger_id`, `outcome_id`, `exception_or_defense_id`, and `primary_reasoning_id`. These identifiers are inferred by the Dasha agent from the Frank packet and the model answer. The code does not decide that an answer is a marriage-provision answer, an estoppel answer, an arbitrary-and-capricious answer, or a plain-meaning answer; it only records and groups the canonical labels that the Dasha agent produced. The cluster key is therefore (`doctrine_id`, `rule_trigger_id`, `outcome_id`, `exception_or_defense_id`, `primary_reasoning_id`).

The fifth tuple element is the normalized primary legal-reasoning family, such as signed-writing compliance, Statute-of-Frauds bar, association replacement control, constructive trust, promissory estoppel, one-year reasoning, plain meaning, or contra proferentem. Dasha stores the full secondary-path audit profile in the cluster artifact, including accepted, rejected, uncertain, and merely mentioned theories, but secondary paths do not automatically split the cluster unless Dasha changes the primary canonical fields. This prevents a common failure mode in legal answers: a response may discuss five possible gates, reject four, and still rest on the same controlling path as another answer. Rejected paths remain inspectable, but they do not by themselves fragment the centroid.

This is important for interpreting the evidence: the reported Dasha configuration does not use embeddings or a trained clustering model. It is an LLM semantic-extraction layer followed by deterministic grouping over agent-emitted canonical identifiers. That design is transparent and auditable, but it means the quality of clustering depends on whether Dasha correctly extracts the legal abstraction from the source context and answer.

Responses with the same normalized bucket tuple are grouped into clusters. Each cluster has a representative centroid, member ids, centroid/member consistency metrics, and a legal signal summary. The representative response is selected by maximizing a combination of token-level Jaccard similarity and feature-match similarity against other members. Mean text similarity is retained as

a warning signal, not as the clustering key, because legally equivalent answers often use different surface language. The run bundle also stores each member’s normalized signature key and the cluster’s normalized key, allowing the validation layer to audit whether every member remains aligned with its centroid. Legacy keyword-based normalization remains only as an offline regression fallback for old fixtures that lack Dasha canonical identifiers; it is not the claimed live research method.

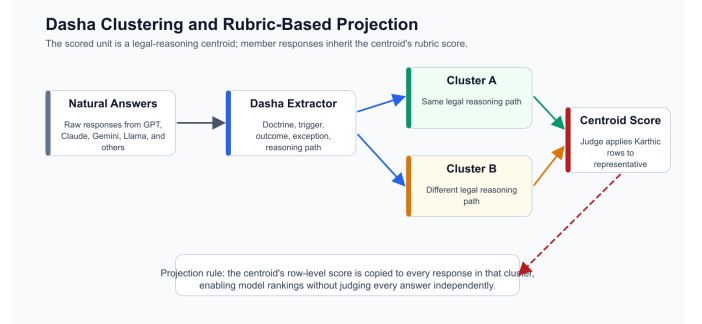


Fig. 3. Dasha clustering and rubric-score projection. Raw model answers are first abstracted into legal-reasoning signatures, grouped into centroids, scored against Karthic rubric rows, and then projected back to all member responses for model-level rankings.

J. Judge and Zak

The judge applies every Karthic rubric row to each Dasha centroid and returns a 0–4 score with rationale. The centroid’s weighted score is projected to all responses in the cluster, enabling model-level rankings. This follows the rubric-conditioned LLM-as-judge pattern in prior work while avoiding pairwise comparisons across every individual response. Zak receives judge and cluster signals and emits an escalation packet only when configured uncertainty, disagreement, or quality failures require further review.

The judge receives the cluster legal signal, the representative response, and a compact copy of every Karthic row. It must return one row score for every rubric row. Scores are clipped to the 0–4 range and combined using the rubric weights. The resulting centroid score is copied to every response in that cluster. Model rankings are then computed as the mean projected score for each model identifier. The live judge configuration now supports repeated judge calls and judge panels. When repeats or multiple judge models are enabled, each centroid-row score is collected multiple times and accompanied by a stability report containing pairwise mean absolute error, pairwise weighted kappa, maximum row-score range, unstable row flags, and the recorded judge-panel composition. Rows whose repeated or panel scores differ by two or more scale points are sent to an adjudicator, which receives the prior scores, the rubric, the cluster signal, and the representative answer, then emits the final row score. Zak still records the disputed cluster state rather than letting adjudication

erase provenance. This projection rule is why a run with one Dasha cluster can produce tied model rankings: if all models use the same legal reasoning path, the method treats the centroid as the scored unit. Zak is intentionally simple in the reported implementation: if judge agreement, judge-repeat stability, or quality thresholds fail, it emits a packet containing the disputed cluster ids and rubric row ids; otherwise it records that no escalation was required.

V. Experimental Design and Validation

The validation protocol tests whether each automated stage produces a source-grounded artifact that is usable by the next stage without manual patching. The protocol is internal: it measures reproducibility, schema integrity, perturbation behavior, cluster coherence, judge stability, and artifact quality before external legal-expert review. A generated run bundle is the unit of analysis. It contains the Frank packet, Karthic rubric, response artifacts, Dasha clusters, judge scores, Zak packets, manifest, validation summary, tables, and prompt-provenance records.

The internal validator checks five stage families, and the paper separately reports a stage-level artifact assessment. Frank passes when the packet contains a doctrine, source-grounded question, variations, gold answer, and a complete downstream handoff record. Karthic passes when the rubric has enough rows, canonical categories, non-duplicated criteria, and source-support snippets. Dasha has two separate gates. The coverage gate passes when unlabeled model answers to the same Frank question are all assigned to clusters after response generation, with no preassigned reasoning labels used as input. The divergence gate passes only when the configured live run observes at least the required number of distinct legal-reasoning clusters. Judge passes when every cluster receives every rubric row score, model rankings are generated, and any repeated-judge stability report does not require review. Zak passes when escalation packet state is explicitly captured, including escalation for unstable repeated judge rows when applicable. Perturbation validation passes when the run contains both invariant and material question edits, invariant edits preserve the dominant legal answer path, and material edits change the dominant answer path or reasoning signature. This is a domain-specific metamorphic test: the validation target is the relation between the original and edited legal questions, not a standalone answer label for each question.

The validation logic is implemented as executable checks rather than prose-only review. Frank checks target packet completeness and deprecated-heading failure modes. Karthic checks row count, required category coverage, duplicate-like row criteria, source-support fields, and weight normalization. Dasha checks that all responses are assigned to clusters, that clusters are internally consistent under the same normalized signature-bucket function used by the clustering step, and that each member’s normalized signature key agrees with its centroid key. It also checks

that no unexpected non-Latin text appears in legal signals. Judge checks that the number of returned row scores equals clusters times rubric rows, that every model receives projected member scores, and that rankings are materialized. When judge repeats or judge panels are configured, it also checks the stability summary produced from repeated or panel row scores and records the panel composition. Zak checks explicit escalation state even when no packet is needed. These checks were chosen because they correspond to the pipeline’s main failure modes: unsupported legal framing, generic rubrics, schema drift, paraphrase over-splitting, missing judge rows, and silent escalation failure.

The statistical layer adds uncertainty estimates to those executable gates. The implementation reports Wilson intervals for binomial rates such as member/centroid coherence and unstable judge rows; bootstrap intervals for cluster sizes, feature similarity, and projected model scores; permutation tests over categorical answer-family distributions for perturbation tracks; pairwise mean absolute error and weighted κ for judge-panel agreement; and macro-F1, purity, and completeness for the controlled 500-response regression. These statistics are reported as diagnostics of the method under the current run design. They are not used to claim population-level model superiority from one case.

The evidence package has five layers. First, unit and regression tests exercise config loading, prompt hashing, Frank packet completeness, Karthic rubric coverage, Dasha clustering invariants, judge projection, repeated-judge and panel-judge stability aggregation, Zak escalation, model-call clients, and statistical metrics. Second, a real live batch executes the full LLM-driven path using ten response model identifiers with two samples per model over three question tracks, yielding 60 natural answers. Third, a natural-response Dasha audit checks that the clustered inputs are unlabeled model answers to one Frank question, not responses generated from preassigned reasoning categories. It also checks whether the batch produced enough observed reasoning divergence to stress Dasha and whether each clustered member agrees with its centroid key. Fourth, a deterministic 500-response regression suite tests the clustering container on eight known Statute-of-Frauds signatures; that suite is a scale and bookkeeping check, not the main evidence for Dasha’s discovery behavior. Fifth, a perturbation smoke test executes an original question, an invariant party-name edit, and a material no-writing edit, then verifies that the invariant edit preserves the answer path while the material edit changes it.

The validation is intentionally layered. Unit tests isolate parser and invariant logic without paying for model calls. The natural-response batch then exercises the live path that matters scientifically: multiple model outputs answer the same source-derived legal question before Dasha sees them. The reported run passes the configured Dasha divergence gate because it produced 23 track-aware ob-

served reasoning clusters against a target of two, but it remains a bounded single-case sample. The deterministic 500-response regression is retained because it catches implementation regressions in clustering, projection, and table generation at the target response scale, but it is not used to claim that Dasha has discovered reasoning families from unlabeled model text. This distinction is central to the method: discovery is tested only on unlabeled model answers; labeled fixtures test software bookkeeping. Separate Zak regressions force low-margin and unstable-judge conditions so that escalation packet creation is tested as a positive behavior, not only as an empty no-escalation record.

A Dasha-specific refinement tests a multi-path legal-answer failure mode. Using the same 60 saved natural responses, Dasha signatures were regenerated with a schema that separates the controlling path from secondary paths. A naive cluster key that included every secondary path produced 49 clusters for 60 responses, which was too fragmented for centroid judging. The retained method therefore stores the full secondary-path audit profile but clusters on Dasha’s primary canonical fields. The tuned track-aware pass produced 23 clusters across the original, invariant, and material tracks after repairing a normalization bug around negative enforceability language. Judge was then rerun on those multi-path clusters, producing 230 final row-level scores, projected scores for all 60 member responses, and updated model rankings.

The internal review layer then asks the legal-methodological questions that schema checks cannot answer by themselves: whether Frank’s framing is what a researcher expected from the source and instruction examples; whether Karthic’s rows are legally useful rather than generic; whether Dasha’s sampled cluster members share a legal theory with their centroid; and whether the judge rationales follow the rubric rows. The manuscript therefore reports both executable gates and human-readable artifact examples. The executable gates make the run reproducible; the artifact assessment makes the legal quality claim inspectable.

This protocol does not replace expert legal review. It is the complete internal evidence gate before any external claim: the pipeline must prove that it can run, preserve provenance, expose errors, cluster at controlled scale, and generate coherent artifacts before any stronger publication claim is made.

VI. Experiments and Results

A. Live Source-to-Ranking Run

The claim-supporting source input is a real court-case PDF rather than a synthetic hypothetical or manually written fact summary. The Statute-of-Frauds calibration case is *Anglemire v. Policemen’s Benevolent Association of Chicago*, 301 Ill. App. 277, 22 N.E.2d 713 (Ill. App. Ct. 1939). The run configuration points to the stored PDF under cases/MarriageSoF_Anglemire v Policemens

Benev Assn of Chicago.pdf; the protocol-freeze artifact records the PDF content hash, extracted-text hash, case id, citation, jurisdiction, court, decision date, and source limitations. This matters methodologically because Frank must derive the benchmark from a citable source authority, not from a reconstructed scenario supplied by the researcher.

The live LLM-driven pipeline was run on that real Statute of Frauds calibration case using ten response model identifiers with two samples per question track: gpt-5.2, gpt-5.4, claude-sonnet-4-6, claude-haiku-4-5-20251001, google/gemini-3-flash, google/gemini-3-pro, anthropic/claude-4.5-sonnet, openai/gpt-4.1-mini, deepseek-ai/deepseek-r1, and meta/meta-llama-3-70b-instruct. Frank and Karthic used gpt-5.2. Dasha used claude-sonnet-4-6 to extract and canonicalize legal-reasoning signatures, and Judge used a two-model claude-sonnet-4-6/claude-haiku-4-5-20251001 panel with claude-sonnet-4-6 adjudication for unstable rows.

The run used the corrected natural response protocol: each response model received only the Frank-generated question, with no system prompt, no hidden gold answer, and no required memo headings. The run produced 60 model responses over the original, invariant, and material question tracks; 10 rubric rows; 23 Dasha clusters; 230 final row-level judge scores; adjudicated final centroid scores; 60 projected member scores; and 10 model ranking entries. Table III summarizes the resulting stage checks.

TABLE III
Internal validation summary for the live natural-response batch.

Stage	Status	Evidence
Frank	pass	doctrine=Statute of Frauds — agreements made upo
Karthic	pass	rows=10; categories=conclusion,counterargument,doc
Dasha	pass	responses=60; clusters=23; min_observed_clusters=
Judge	pass	judge_model=claude-sonnet-4-6,claude-haiku-4-5-20
Zak	pass	needs_zak=True; packets=1
Perturbations	pass	tracks=3; invariant_checks=1; material_checks=1; s

B. Statistical Validation Diagnostics

Table IV reports the statistical diagnostics that are computable from the current run bundle. These diagnostics do not convert the single-case study into a population-level benchmark. Instead, they quantify uncertainty for the specific methodological claims made in this paper: Dasha member/centroid coherence, observed reasoning diversity, perturbation behavior, judge-panel stability, projected-ranking uncertainty, and controlled 500-response regression behavior.

The diagnostics sharpen the interpretation of the evidence. Dasha assigned all 60 live responses to centroids without a member/signature mismatch (Wilson 95% CI: 0.940–1.000), which supports the cluster bookkeeping claim. The material signed-writing perturbation produced a large response-distribution shift from the original track

TABLE IV
Statistical validation diagnostics for the live run and controlled regression.

Validation target	Estimate	Uncertainty/statistic	Interpretation
Dasha member coherence	1.000	95 percent Wilson CI [0.940, 1.000]	No centroid/member key mismatches in the saved run; validates bookkeeping, not expert legal truth.
Dasha cluster diversity	23 clusters; effective count 2.71	Mean size 2.61; bootstrap CI [1.61, 3.87]	Natural responses did not collapse into a single reasoning family.
Judge panel stability	weighted kappa 0.630; MAE 0.465	unstable rows 18/230; Wilson CI [0.050, 0.120]	Adjudication is required and recorded; judge scores are not treated as silently final.
Projected ranking uncertainty	gpt-5.2	mean 3.080; bootstrap CI [2.010, 3.950]; top-two overlap=True	The live ranking is inspectable but not statistically decisive across model families.
Controlled 500-response regression	macro-F1 1.000; purity 1.000	n=500; clusters=8	Confirms scale/bookkeeping under known signatures; not natural discovery evidence.
Perturbation: V1	TVD 1.000	permutation p=0.0005; passed=True	material edit: base dominant=oral_promise_void_later_beneficiaries_p track dominant=writing_satisfied_promise_enforceable
Perturbation: V_surface_invariant	TVD 0.100	permutation p=1.0000; passed=True	invariant edit: base dominant=oral_promise_void_later_beneficiaries_p track dominant=oral_promise_void_later_beneficiaries_p

(total variation distance 1.000; permutation $p = 0.0005$), while the invariant party-name edit preserved the dominant answer family (total variation distance 0.100; permutation $p = 1.0000$). The judge panel showed moderate row-level agreement before adjudication (weighted $\kappa = 0.630$, mean absolute error 0.465), and 18 of 230 row scores required adjudication. Projected model-score intervals overlap near the top of the ranking, so the ranking is an auditable pipeline output, not a statistically decisive model-family comparison.

This live run is intentionally bounded to control cost. Its purpose is not to estimate population-level model performance with narrow confidence intervals. It demonstrates that the real model-generation path runs from source case to model ranking without manual patching, and that Dasha can separate multiple observed legal-reasoning paths in a perturbation-aware natural-response batch. The appendix therefore includes direct run-bundle excerpts for every stage: Frank’s generated scenario, gold answer, and perturbations; Karthic’s complete row-level rubric; natural model responses; Dasha cluster membership and reasoning signatures; Judge row-score rationales; and the Zak escalation packet. These examples are generated from the same artifacts used to compute the tables, so the manuscript’s practical claim can be inspected at the artifact level rather than inferred only from aggregate metrics.

C. Natural-Response Dasha Audit

The relevant Dasha test is whether responses are clustered after models answer the same Frank-generated question. The live audit therefore checks the completed run bundle for unlabeled model answers, the expected Frank

question tracks, full cluster coverage, and no preassigned reasoning labels. Table V summarizes that audit.

TABLE V
Natural-response Dasha audit for the live model batch.

Check	Value
Unlabeled model responses	60
Distinct response models	10
Frank questions answered	3
Preassigned reasoning labels	0
Observed Dasha clusters	23
Member/centroid coherence	1.0
Minimum target clusters	3
Divergence status	observed_reasoning_divergence
Clustered responses	60
Audit status	natural_response_audit_passed

This audit is intentionally separate from the deterministic stress suite. It is the evidence that Dasha is operating on model outputs rather than on categories given to the response generator. In this batch, the normalized Dasha signatures formed 23 track-aware clusters. On the original and invariant tracks, the dominant cluster family rejected enforcement of the unwritten marriage- consideration promise and let the later bylaw-compliant designation control. Minority clusters captured equitable override, constructive-trust, and certificate-as-writing theories. On the material signed-writing track, the dominant answer family shifted toward enforceability or written-compliance reasoning. The audit therefore passes the configured divergence and perturbation gates, while still remaining too small to support broad claims about all models or all Statute-of-Frauds cases.

TABLE VI
Dasha cluster summary for the live natural-response run.

Cluster	Size	Track
original__cluster_1	11	original
original__cluster_2	4	original
original__cluster_3	1	original
original__cluster_4	1	original
original__cluster_5	1	original
original__cluster_6	1	original
original__cluster_7	1	original
V_surface_invariant__cluster_1	1	V_surface_invariant
V_surface_invariant__cluster_2	10	V_surface_invariant
V_surface_invariant__cluster_3	6	V_surface_invariant
V_surface_invariant__cluster_4	1	V_surface_invariant
V_surface_invariant__cluster_5	1	V_surface_invariant
V_surface_invariant__cluster_6	1	V_surface_invariant
V1__cluster_1	1	V1
V1__cluster_2	3	V1
V1__cluster_3	1	V1
V1__cluster_4	4	V1
V1__cluster_5	1	V1
V1__cluster_6	1	V1
V1__cluster_7	2	V1
V1__cluster_8	2	V1
V1__cluster_9	1	V1
V1__cluster_10	4	V1

TABLE VII
Projected model rankings from centroid scores.

Model	Mean projected score	ResponseF1
gpt-5.2	3.080	6
google/gemini-3-flash	3.077	6
gpt-5.4	3.027	6
claude-sonnet-4-6	2.867	6
openai/gpt-4.1-mini	2.853	6
deepseek-ai/deepseek-r1	2.853	6
google/gemini-3-pro	2.767	6
claude-haiku-4-5-20251001	2.463	6
anthropic/claude-4.5-sonnet	2.400	6
meta/meta-llama-3-70b-instruct	0.457	6

D. Dasha Multi-Path Audit

The live Anglemire responses also exposed a Dasha-specific representation issue: natural legal answers often discuss several gates or equitable theories before settling on one. Regenerating Dasha signatures with explicit primary-path and secondary-path fields completed for all 60 saved responses without malformed JSON failures. A first pass that clustered on every secondary path overfragmented, so the tuned rule preserves the full secondary-path audit profile while clustering on Dasha’s primary canonical fields. After repairing a negative-language normalization error around words such as “unenforceable,” the full source-to-score rerun produced 23 track-aware clusters: 7 on the original question, 6 on the surface-invariant edit, and 10 on the material signed-writing edit. Judge scored those 23 centroids with a two-model panel, yielding 46 panel calls, 230 final row-level scores, 11 adjudications, projected member scores for all 60 responses, and updated model rankings. The most common cluster profile

remained no material secondary path, but the audit also captured uncertain promissory-estoppel or reliance analysis, accepted part performance analysis, accepted reliance analysis, and accepted signed-writing analysis.

Score	Normalized reasoning key
3.96	oral_promise_void_later_beneficiaries_prevail / part_performance_estoppel_raised_reliance
3.90	oral_promise_void_later_beneficiaries_prevail / part_performance_estoppel_raised_reliance
3.18	oral_promise_void_later_beneficiaries_prevail / promissory_estoppel_raised_reliance
0.20	promissory_estoppel_enforced / part_performance_estoppel_raised_reliance
0.30	promissory_estoppel_enforced / promissory_estoppel_raised_reliance
2.30	sof_bars_contract_estoppel_uncertain / part_performance_estoppel_raised_reliance
3.08	sof_bars_contract_estoppel_uncertain / promissory_estoppel_raised_reliance
0.00	sof_bars_contract_estoppel_uncertain / part_performance_estoppel_raised_reliance
3.96	oral_promise_void_later_beneficiaries_prevail / part_performance_estoppel_raised_reliance
3.96	oral_promise_void_later_beneficiaries_prevail / part_performance_estoppel_raised_reliance
0.30	promissory_estoppel_enforced / part_performance_estoppel_raised_reliance
0.16	writing_satisfied_promise_enforceable / executed_contract_vested_equitable
0.56	writing_satisfied_promise_enforceable / bylaw_affidavit_procedure_val
0.72	writing_satisfied_promise_enforceable / bylaw_change_right_no_irrevocable
1.34	writing_satisfied_promise_enforceable / executed_contract_vested_equitable
0.56	writing_satisfied_promise_enforceable / part_performance_estoppel_raised_reliance
0.88	writing_satisfied_promise_enforceable / promissory_estoppel_raised_reliance
0.60	writing_satisfied_promise_enforceable / bylaw_change_right_no_irrevocable
1.02	writing_satisfied_promise_enforceable / bylaw_change_right_no_irrevocable
1.98	writing_satisfied_promise_enforceable / part_performance_estoppel_raised_reliance
0.72	writing_satisfied_promise_enforceable / writing_satisfies_sof / writing_satisfies_sof
V_surface_invariant	invariant_invariant_preserved
V1	material_material_difference_observed
	pass

F. Controlled 500-Response Dasha Regression Test

The 500-response suite is now treated as a deterministic regression test rather than a robustness demonstration. It generates fixture answers across eight legally distinct Statute-of-Frauds reasoning archetypes so the software can test cluster bookkeeping, centroid projection, table generation, and metrics at the target scale. It does not show that Dasha can discover reasoning families from natural model outputs.

TABLE IX
Controlled Dasha clustering regression test.

Check	Value
Responses	500
Expected reasoning archetypes	8
Observed Dasha clusters	8
Cluster purity	1.0
Cluster completeness	1.0
Macro-F1	1.0
Mean cluster size	62.5

The regression test produced eight observed clusters for eight expected reasoning archetypes. Cluster purity, cluster completeness, and macro-F1 were all 1.0. This proves that the clustering layer preserves normalized legal-reasoning distinctions at the target 500-response scale when the signatures are already known. It does not substitute for the natural-response audit; instead, it functions as a regression and scale test for cluster bookkeeping, centroid projection, and metrics generation.

G. Implementation Test Coverage

The reference implementation has 82 research-pipeline tests covering offline run-bundle exports, Frank packet quality, Karthic rubric quality, SOF gate regression cases, Dasha clustering invariants, sparse-response gate context, LLM reasoning signature clustering, source-derived Dasha gate aliasing, a non-SOF contract-interpretation transfer smoke, judge projection and model ranking, low-margin Zak escalation, repeated-judge and panel-judge stability aggregation, Dasha member/centroid audit generation, protocol-freeze manifests, manuscript reference and hygiene linting, model roster config loading, model-call polling, natural-response audit generation, live-config preflight, credential-alias checks, live-call budget caps, runtime budget enforcement, run-bundle integrity auditing, secret and local-path linting, internal review packet generation, claim-ledger generation, handoff-manifest generation, consolidated no-call audit, method-readiness reporting, review-readiness table generation, metrics, perturbation-track generation, perturbation reporting, and internal stress-table generation. These tests make the pre-expert-review manuscript reproducible: the reported claims are backed by executable checks rather than by visual inspection alone.

VII. Qualitative Artifact Analysis

Executable validation checks do not by themselves establish that a legal evaluation artifact is useful. We therefore analyze the reported Statute-of-Frauds run as a source-to-score case study grounded in the real court case *Anglemire v. Policemen’s Benevolent Association of Chicago*. The analysis follows the same artifact boundaries used by the implementation: Frank, Karthic, natural response generation, Dasha clustering, rubric-conditioned judging, Zak escalation, and paper export. At each boundary, we ask whether the artifact is legally coherent, source-grounded, consistent with the role instructions, and auditable by an independent research team.

For Frank, the central question is whether the source case has been converted into a legally sound benchmark rather than merely summarized. In the reported run, Frank identified the Statute-of-Frauds marriage-provision theory and its interaction with fraternal-benefit certificate rules; generated a neutral Illinois-law question; produced a structured gold answer; and created three boundary variations. In the perturbation lane, those variations become executable question tracks rather than prose-only notes. The artifact is useful for technical review because it preserves the legally operative facts: the premarital promise, marriage consideration, certificate naming the wife, later replacement certificate, and association rules. The remaining limitation is that the legal assessment is still internal: the evidence supports the claim that Frank produced a coherent source-grounded packet, not that outside experts have certified the gold answer.

TABLE X
Stage-level artifact assessment for the live SOF case study.

Stage	Validation question	Current evidence	SOF	Internal assessment
Frank	Is the source converted into a legally sound gold packet and neutral benchmark question?	Identified marriage-provision and fraternal-benefit certificate issues; generated one neutral question, structured gold answer, and 3 variations.		Coherent source-grounded packet for the case; requires external legal review before publication.
Karthic	Is the rubric fresh, comprehensive, source-supported, and derived from Frank?	Produced 10 rows across doctrine, rule, facts, exceptions, counterargument, conclusion, variation, and source-support categories.		Good scoring core for internal validation; full row-card materialization remains a publication-hardening task.
Dasha	Are natural model answers clustered by legal reasoning, not labels or surface form?	Clustered 60 unlabeled natural responses into 23 track-aware reasoning clusters with all responses assigned.		Meaningful perturbation-aware separation; needs held-out legal cases before broad validity claims.
Judge	Is the rubric applied row by row to centroids and projected to members?	Produced 230 row-level scores, 60 projected member scores, and 10 model ranking entries.		Mechanism works; ranking is not yet a population-level model-performance claim.
Zak	Are failures and uncertainty made explicit?	The live SOF run emitted 1 Zak packet after judge disagreement/adjudication checks.		Packet creation works in the active live run; live threshold calibration remains.

For Karthic, the question is whether the rubric is a fresh scoring instrument derived from Frank rather than a generic grading checklist. The reported run produced 10 rows covering doctrine, rule, facts, exceptions, counterargument, conclusion, variation sensitivity, and source support. That is the right shape for the reported case because the answer space turns on doctrinal gate selection, writing or certificate significance, later-beneficiary defenses, equitable theories, and disciplined conclusion language. The reference implementation stores the scoring core of each row rather than every full row-card field specified in the ideal Karthic instructions. This is acceptable for the present reported run, but the manuscript marks it as a limitation before publication.

For Dasha, the question is whether unlabeled natural model answers are grouped by legal reasoning. The reported run used 60 natural responses across the original question, an invariant party-name edit, and a material signed-writing edit. It observed 23 track-aware clusters after the multi-path Dasha repair. The dominant original and invariant families rejected the unwritten marriage-consideration promise and treated the later bylaw-compliant designation as controlling; minority original clusters captured equitable override, constructive-trust, and certificate-as-writing theories. The material signed-writing track shifted the dominant answer family toward written compliance and enforceability. This is the kind of separation the pipeline is intended to find, because the clusters differ in legal theory rather than merely in wording. For perturbation runs, Dasha clusters each question track separately before comparing dominant signals, which prevents a variant answer from being absorbed into the original question’s cluster. The strongest remaining risk is extraction error: Dasha in the reported configuration uses an LLM to extract a reasoning signature, then groups exact normalized buckets. The method is auditable, but future validation should repeat extraction, compare alternate clusterers, and increase the number of natural model responses.

For Judge, the question is whether the rubric is actually applied row by row to the Dasha centroids. The reported run generated 230 final row scores: 10 rubric rows across 23 clusters judged by a two-model panel. Eleven clusters with unstable panel rows received an adjudication pass before final projection. The judge then projected centroid scores to all 60 member responses and produced model rankings. This demonstrates the intended compression mechanism: score the legal-reasoning centroid, then propagate the score to responses using that reasoning path. The present ranking is not a population-level claim about model quality because the live batch is bounded and uses one source case. It is evidence that the judge machinery can apply the rubric, adjudicate unstable panel rows, and produce auditable rankings from centroid scores.

For Zak, the validation question is whether uncertainty is represented explicitly. In this run, the judge output required escalation after centroid scoring and adjudication, and Zak emitted a disputed-cluster packet rather than silently treating the ranking as final. The implementation also includes positive offline regressions: one forces a low-margin centroid comparison to create a disputed-cluster packet, and one forces unstable repeated judge rows to create an unstable-row packet. These regressions show that Zak packet creation is not merely an empty no-escalation record. The remaining work is to calibrate the live thresholds across held-out cases so escalation is neither too rare nor too noisy.

A. Doctrine Transfer

The pipeline is designed to be doctrine-general, but the evidence in this manuscript is intentionally narrower. The doctrine-transfer claim is therefore architectural rather than empirical: the same source-to-packet schema, instruction-context mechanism, natural-response protocol, reasoning-signature extraction, centroid judging, and escalation logic can be reused for another area of law if that area supplies adequate source material and role-specific instruction context. Statute of Frauds is the calibration domain because it contains multiple gates, exceptions, boundary variations, and plausible model failure modes. Those properties make it a useful stressor for the general pipeline design.

TABLE XI
Doctrine-transfer argument and remaining validation burden.

Pipeline component	Doctrine-general mechanism	Remaining validation burden
Frank	Uses source text plus instruction context to infer doctrine gates and create a packet schema that is not inherently SOF-only. An offline contract-interpretation smoke fixture now emits generic doctrine gates without a SOF packet.	Run live held-out non-SOF cases and review doctrine detection, question neutrality, and gold-answer quality.
Karthic	Builds rows from the locked Frank packet, so rubric content should follow the source doctrine rather than a fixed SOF checklist. The contract fixture produces a generic source-grounded rubric.	Confirm row coverage and source support in additional doctrine families.
Dasha	Extracts normalized reasoning signatures from natural answers after generation and can use Frank’s source-derived gate aliases. The contract fixture separates buyer-remedy, contra-proferentem, and seller-limitation reasoning.	Test whether signature fields and buckets remain discriminative outside SOF on live natural model answers.
Judge	Applies row-level rubric criteria to centroids, independent of the doctrine label.	Repeat judging with alternate judge models, repeats, and expert comparison.
Zak	Escalates low agreement, mixed clusters, or uncertain source support using artifact metadata.	Add positive failure cases across domains to verify escalation sensitivity.

The current implementation also includes an offline contract-interpretation transfer smoke test. That fixture verifies that the artifact schema and Dasha normalization

path can run outside Statute of Frauds, but it is intentionally reported as schema-transfer evidence rather than broad legal-domain validity. The strongest future claim would require held-out live cases from other legal domains. For each domain, the same stage-level questions should be asked: did Frank infer the correct doctrinal elements from the source; did Karthic build a useful rubric from that packet; did Dasha recover and cluster natural model reasoning paths; did Judge apply the rubric consistently; and did Zak surface failures instead of hiding them? Until that evidence exists, the paper should claim that the method is designed for doctrine transfer, internally demonstrated on Statute of Frauds, and smoke-tested on one non-SOF fixture, not that broad legal-domain validity has already been proven.

VIII. Reproducibility Specification

This section states the implementation contract independently of any particular storage layout. A researcher can implement the method in any programming environment if the implementation preserves the inputs, artifacts, normalization rules, and validation checks described below.

A. Run Configuration

Each run is defined by a frozen configuration. The configuration must include the legal source text or source identifier, a run identifier, the response-model roster, sample count per model, stage-specific temperature policy, perturbation policy, clustering method, judge model, quality gates, and output location. The temperature policy must distinguish deterministic benchmark construction from stochastic response generation: Frank and Karthic may be fixed at temperature 0 for reproducible source and rubric artifacts, while response models should use nonzero sampling temperatures and repeated samples when the goal is to observe natural variation in legal reasoning. Judge calls should record their temperature and, for reliability claims, use repeats or a panel so score variability is measured. The configuration should also record whether the run is offline, live, or a fixture stress test. All stochastic choices, including model temperature offsets and fixture sampling seeds, must be explicit.

The reference implementation materializes this contract with a protocol-freeze manifest before a run is treated as frozen. The freeze manifest records the configuration hash, source hash, ordered instruction-stream hashes for Frank, Karthic, and Dasha, response-model roster, clustering settings, judge model or judge-panel roster, perturbation policy, quality gates, normalization versions, and a top-level protocol hash. It intentionally omits API keys and generated model outputs. This separates the research protocol from mutable local environment state and makes later result bundles auditable against the exact method that was intended to run.

The repository also includes a static manuscript linter for environments without a TeX engine. The linter does not replace latexmk or pdflatex, but it verifies that all `\input` targets, figure files, bibliography files, and citation keys referenced by the paper resolve. A compiled PDF remains the stronger artifact when a full TeX installation is available.

Finally, the implementation provides a no-call preflight check for live configs. Preflight builds the protocol-freeze metadata in memory and checks that the run is live-capable, uses natural response prompting, has a sufficiently diverse response roster, uses LLM agents for Frank, Karthic, and Dasha, uses reasoning-signature clustering, requires multiple observed clusters, configures LLM judging with repeats or a panel, and can find the source case. It also reports local credential and perturbation-policy warnings before any model call is made. This makes budgeted live runs auditable before execution rather than only after artifacts have been generated.

TABLE XII
Minimum run-configuration fields required to reproduce a pipeline run.

Field group	Required contents
Source	Source-case text or stable source identifier; jurisdiction if the benchmark depends on jurisdiction-specific law; source hash.
Agents	Model identifier, provider adapter, temperature, instruction-stream hash, and output schema for Frank, Karthic, Dasha extraction, and Judge.
Responses	Response-model identifiers, sample count per model, temperature policy, response prompt style, and question track ids.
Perturbations	Whether perturbations are enabled; maximum number of variations; requirement for invariant and material tracks; expected behavior for each track.
Clustering	Signature schema version; normalization version; minimum observed clusters; mixed-cluster threshold; representative-selection rule.
Judge and Zak	Rubric scale, judge aggregation rule, disagreement threshold, escalation margin, and packet-generation rule.
Outputs	Versioned JSON artifacts, prompt hashes, response provenance, validation summary, generated tables, and figure inputs.

B. Artifact Schemas

The pipeline is reproducible because stages exchange versioned artifacts rather than conversational state. Frank emits the benchmark packet. Karthic emits rubric rows. Response generation emits raw model answers with question-track metadata. Dasha emits clusters with member ids and legal signals. Judge emits row-level centroid scores and projected member scores. Zak emits either an empty no-escalation state or an escalation packet.

TABLE XIII

Required artifact interfaces for an independent implementation.

Artifact	Required fields
Frank packet	id, source_hash, doctrine_family, detected_doctrine_gates, source_extraction, neutral_question, gold_answer, variations, controller_card, prompt_hashes.
Karthic rubric	id, source_hash, frank_packet_id, rows, weights, prompt_hashes. Each row requires an id, category, weight, criterion, source-support field, and scoring anchors or equivalent rubric guidance.
Responses	id, model, sample_index, text, question_id, track_id, variant_id, perturbation_type, expected_behavior, timestamp or generation provenance.
Dasha clusters	id, method, legal_signal, representative_response_id, member_response_ids, size, centroid_quality, and, in perturbation runs, question-track metadata.
Judge scores	One row-level score and rationale for every rubric row and every cluster representative; weighted centroid scores; projected member scores; model rankings.
Zak packets	Escalation status, triggering threshold, disputed cluster ids, disputed rubric row ids, rationale, and handoff text when escalation is required.

C. Instruction-Stream Construction

The agent context is an ordered prompt bundle, not hidden memory. A reproducible implementation should store each prompt component with a stable component id, role, version, and text. At runtime, the implementation concatenates the components in fixed order, prefixes each component with its id, truncates only at a declared token or character budget, and records a cryptographic hash over the rendered stream. The same rendered stream and upstream artifact must be supplied to the same model identifier to reproduce the stage input.

The shared first component is a cross-agent legal reasoning protocol. It states that source authority controls legal substance, that calibration examples do not override the source, that schema fields are doctrine-general, that uncertainty is preserved rather than silently filled, and that perturbations must identify the expected legal effect of the edited fact. Role-specific components then define the agent contract. Frank receives source-intake, packet-building, question writing, gold-answer, and variation instructions. Karthic receives packet-audit, row-card, weighting, duplicate-control, and scoring-overlay instructions. Dasha receives signature-extraction, track-audit, clustering, centroid-metadata, and escalation instructions. Judge receives rubric-application and row-score rationale instructions.

D. Agent Calls

Frank receives the source text and its instruction stream, then returns one strict JSON packet. Karthic receives the locked Frank packet and its instruction stream, then returns one strict JSON rubric. Response models

receive only the question text for the active track under the natural response protocol; they do not receive the gold answer, rubric, hidden reasoning labels, or required memo headings. Dasha receives the locked Frank packet, its instruction stream, and one raw response at a time, then returns a strict JSON reasoning signature. Judge receives one Dasha representative, the cluster legal signal, and all Karthic row criteria, then returns one score and rationale per row.

E. Dasha Clustering Algorithm

The reported configuration clusters responses by Dasha-extracted legal-reasoning signatures rather than by surface text. For each response, Dasha emits both natural-language fields and canonical identifiers: `doctrine_id`, `rule_trigger_id`, `outcome_id`, `exception_or_defense_id`, and `primary_reasoning_id`. The identifier vocabulary is not pre-enumerated in code. Dasha derives it from the Frank packet, source-derived gates, instruction context, and response text. The implementation sanitizes the signature, removes unexpected script fragments, collapses whitespace, and groups exact canonical identifiers.

The cluster key is the tuple (`doctrine_id`, `rule_trigger_id`, `outcome_id`, `exception_or_defense_id`, `primary_reasoning_id`). These values are Dasha’s canonical legal abstraction. Secondary paths remain in the audit profile so reviewers can see accepted, rejected, and uncertain side theories, but they do not automatically split clusters unless Dasha changes one of the primary canonical fields. Legacy keyword buckets exist only for archived fixtures without Dasha canonical identifiers.

Responses are grouped by exact cluster-key match within each question track. A perturbation run never clusters different tracks together. For each cluster, the representative response is the member with the highest average similarity to the other members, combining token-level Jaccard similarity with feature agreement over gate, outcome, exception, and reasoning fields. Text similarity is retained as a centroid-quality diagnostic, not as the clustering key. A cluster is flagged for review when sampled members visibly use different legal theories, when the normalized signal conflicts with the response text, or when centroid/member feature agreement falls below the configured threshold.

F. Score Projection and Ranking

The judge scores only cluster representatives. Each representative receives a 0–4 score and rationale for every Karthic rubric row. The weighted centroid score is then copied to every response assigned to that centroid. Model rankings are computed from the projected member scores, typically by averaging all projected scores for a model across samples and question tracks. This projection rule is valid only if Dasha cluster membership is legally coherent; therefore cluster audit and perturbation validation are prerequisites for interpreting rankings.

IX. Discussion

The central technical idea is to evaluate legal reasoning at the cluster level rather than scoring every response independently. If Dasha can reliably group answers by reasoning path, a judge can score representative centroids and then project those scores to all cluster members. This creates an interpretable middle layer between raw model generations and aggregate model rankings.

The reported evidence supports the feasibility of that design on a small live natural-response batch. The live run shows that the source-to-ranking path works with real model outputs and that Dasha can separate more than one observed legal-reasoning path without preassigned response labels. The controlled 500-response regression test shows that, once reasoning signatures are available, Dasha can preserve eight legally distinct Statute-of-Frauds reasoning paths at the intended compression scale. These two evidence layers answer different questions: the live run tests discovery from natural answers, while the regression suite tests scale, metrics, and bookkeeping under known archetypes.

The stage-level assessment is equally important. Frank and Karthic are not validated merely by emitting JSON. They are validated by whether the generated question, gold answer, variations, and rubric rows preserve the legally operative facts and match the methodology described in the instruction context. Likewise, Dasha is not validated merely by reporting a cluster count; its clusters must be interpretable as different legal theories. Judge is not validated merely by producing a scalar score; it must apply the Karthic rows to the centroid and leave row-level rationales that can be reviewed. This is why the run bundle and paper appendix include concrete artifacts rather than only aggregate metrics.

The broader transfer claim is deliberately modest. The architecture is designed so that the case source and instruction stream determine the doctrine-specific content, while the pipeline mechanics remain stable: source extraction, question generation, perturbation-track generation, rubric construction, natural response generation, reasoning clustering, centroid judging, and escalation. That gives a plausible path to other areas of law, but it is not yet broad empirical validation. The next stage is to run the same stage-level and perturbation review protocol on held-out cases and additional doctrine families without changing the pipeline in between.

The remaining risk is not only the cluster container itself. Dasha depends on an LLM semantic-extraction step; if that step assigns incorrect normalized signatures, deterministic grouping will preserve the wrong abstraction. The judge is also a single rubric-conditioned model in the reported run, so the reported model rankings should be read as pipeline outputs rather than stable comparative claims about the model families. The next methodological improvement is a larger held-out natural-response suite

with repeated signature extraction, judge ensembling, rubric ablations, and adversarial signature audits.

X. Limitations

This manuscript reports internal validation, not expert-reviewed legal validity. The live validation run is bounded and should be interpreted as an internal source-to-ranking validation, not a population-level benchmark of model performance. The natural-response Dasha audit confirms that the clustered inputs are model answers to Frank-generated question tracks rather than responses generated from preassigned labels, and the reported run produced 23 observed track-aware reasoning clusters over 60 answers. That is a meaningful internal Dasha result, but it is still a single-case batch. It does not establish that arbitrary live model outputs will always produce stable or legally correct clusters. The 500-response scale result is a controlled regression test with known reasoning signatures, not a claim that arbitrary live model outputs will always produce perfectly separable clusters. The evidence is therefore strong for pipeline mechanics and internal reproducibility, but still bounded for external legal validity and cross-doctrine generalization.

The Statute-of-Frauds evidence should be read as a strong litmus test, not as an exhaustive doctrinal proof. The live run validates one source-to-ranking case, and the regression suite covers eight major Statute-of-Frauds reasoning archetypes. A claim that the system handles every Statute-of-Frauds fact pattern would require a larger held-out case suite across jurisdictions, transaction types, writing forms, and exception doctrines.

The reported live model roster is broader than a smoke test but still bounded. The manuscript validates the path with ten model identifiers over three tracks; broader model-family claims require larger rosters, more repeated samples, held-out cases, and natural-response batches where multiple models independently converge on or diverge from the same legal reasoning paths. The bootstrap intervals for the current projected rankings overlap near the top of the table, so the present ranking should be read as an output of the source-to-centroid scoring protocol rather than as statistically significant evidence that one frontier model family outperforms another.

The instruction corpus is deepest for Statute of Frauds. The live agents are now given a doctrine-general protocol and must infer doctrine from the source, but other doctrine families will need richer pack-specific examples before external claims should be made about broad legal coverage.

LLM-as-judge methods are known to have bias risks, including position, verbosity, and self-preference biases [1], [5], [6]. The pipeline mitigates some of these risks by using row-level rubrics, source support, centroid-level scoring, a two-model judge panel, and adjudication for unstable panel rows, but future validation should include larger judge ensembles, repeated scoring across held-out cases,

rubric ablation, model-separated judges, and comparison to expert labels.

Finally, the reported Karthic artifact materializes the scoring core of the rubric row card rather than every field described in the instruction spec. The paper is therefore careful to claim automated row-level scoring, source support, category coverage, and non-duplication checks, not full automation of every human-facing row-card annotation that a legal research team may want before external submission.

XI. Validity Boundaries

The evidence in this manuscript supports a methods claim, not a final population-level claim about legal model quality. The strongest supported claim is that the pipeline can run end to end from a real source case, generate all major artifacts without manual patching, preserve provenance, expose perturbation behavior, cluster observed legal-reasoning signatures, apply a row-level rubric judge, project centroid scores, and record escalation state. The unsupported claim would be that the method is already externally validated across legal domains or that the reported model ranking estimates general model quality.

The remaining validity questions are concrete. Frank must be tested on held-out source cases to determine whether it consistently preserves legally controlling facts and avoids answer leakage. Karthic must be compared with expert-authored rubrics to quantify row coverage, source support, and double-counting. Dasha must be evaluated on larger natural-response batches across multiple doctrines and compared with alternative clustering strategies such as embedding-based clustering or human-coded reasoning labels. Judge must be calibrated against expert legal scores to estimate row-level reliability and bias. Zak thresholds must be tuned so escalation captures genuine uncertainty without overwhelming the review process.

The current evidence is therefore sufficient for independent technical review of the method and implementation. It is not sufficient for a broad empirical claim that the pipeline is valid across all legal domains. That broader claim requires held-out source packs, expert annotations, reliability thresholds specified before evaluation, and statistical analysis over cases rather than a single calibration source.

TABLE XIV
Evidence boundaries for the reported validation gates.

Vision gate	Current status	Evidence
Claim-supporting source provenance	Met	run_source=cases/MarriageSoF_Anglemire_v_Policemens_Benev_Assn_of_Chicago.pdf; protocol_source=cases/MarriageSoF_Anglemire_v_Policemens_Benev_Assn_of_Chicago.pdf; case_id=anglemire_v_policemens_benev_dctrine=Statute of Frauds — agreements made upon consideration of marriage (other than the mutual promise to marry); effect on oral promise to name spouse as beneficiary / settle property; question_chars=1512; variations=1
Frank source-to-packet validity	Met	rows=11; categories=conclusion,counterargument,doctrine
Karthic dynamic rubric validity	Met	response_prompt_style=natural; responses=9; models=3
Natural response protocol	Met	bundle_status=run_bundle_reviewable; checks=33; blocking_errors=0
Run bundle integrity	Evidence gap	responses=9; Run larger natural batches across more model families and held-out questions.
Dasha natural-response clustering	Evidence gap	perturbation_report=not_configured; Run invariant and material perturbation tracks with live model responses.
Perturbation validation	Met	judge_model=gpt-5.2; judge_panel=1; row_scores=55
Judge row-level scoring and rankings	Met	needs_zak=True; packets=1
Zak escalation mechanism	Met	stress_status=internal_stress_passed; stress_responses=500; stress_macro_f1=1.0
Controlled scale regression	Partial	preflight_status=live_preflight_passed; warnings=1; Resolve credential warnings before claim-supporting paid live runs.
Live-run preflight	Not yet met	Requires held-out cases, expert review, live perturbation evidence, broader rosters, and frozen reliability thresholds.
Publication readiness		

Appendix A

Appendix: Agent Instruction Context

This appendix expands the instruction-stream contract in implementation-neutral terms. The method does not require a particular repository layout. It requires that each agent receive a bounded, ordered, versioned text context and that the rendered context be hashed and stored with the resulting artifact.

A. Shared Cross-Agent Protocol

All agents begin with the same legal reasoning protocol. The protocol has five rules.

First, source authority controls legal substance. The source case, statute, Restatement section, regulation, contract, or other supplied authority is the controlling legal input for the run. Doctrine examples are drafting aids only.

Second, calibration domains are not global assumptions. The reported validation uses Statute of Frauds because it has useful boundary facts, but the agents must infer doctrine from the source. If the source supports another doctrine, the same packet, rubric, clustering, and judging architecture is used with doctrine-appropriate gates, elements, defenses, and exceptions.

Third, the schema is doctrine-general. Fields such as `doctrine_gates`, `primary_gate_id`, `rule_trigger`, and `exception_or_defense` refer to the controlling decision points of the source-supported doctrine. They are not Statute-of-Frauds-only labels.

Fourth, variations must identify the legal effect of the edited fact. Invariant variations change legally irrelevant surface facts. Material variations change facts that should affect a doctrine element, defense, exception, burden, remedy, or outcome. Each variation records the expected behavior before model answers are generated.

Fifth, uncertainty is preserved. If the source is too thin, jurisdiction-specific, procedurally tangled, or unsupported by the current instruction context, the agent must state the limitation and preserve it for escalation rather than inventing missing law.

B. Frank Context

Frank’s instruction context defines source-to-benchmark generation. It tells Frank to perform source intake, infer the doctrine family, identify material facts and source limits, generate a neutral benchmark question, produce a source-grounded gold answer, create boundary variations, and freeze a controller card for downstream stages. Frank should stop rather than force a benchmark when the source cannot support a stable packet.

The source extraction component records authority level, jurisdiction, procedural posture, legal issue, rule statement, trigger facts, holding or best-supported answer path, limits, source gaps, benchmark confidence, and review flags. The gold-packet component records doctrine family, controlling trigger, gate order, supporting facts, non-trigger facts, competing barriers, exceptions or substitutes, jurisdiction-sensitive points, likely model mistakes, and benchmark posture. The controller card freezes the current question, gold answer, primary gate id, trigger test, trigger facts, non-triggered sibling gates, compliance status, strongest counterargument, fallback theories, omitted control facts, and source metadata.

Frank’s question-writing component controls what response models see. The question must be neutral, exam-like, and source-faithful. It should preserve the holding-driving facts and one or two plausible distractors while avoiding answer leakage. The call of the question should not reveal the controlling doctrine, tell the model which side should win, or turn the prompt into a checklist of subissues. Jurisdiction is stated only when the benchmark depends on jurisdiction-specific law.

C. Karthic Context

Karthic’s instruction context defines rubric construction from the locked Frank packet. Karthic is not asked to create a generic grading sheet. It builds a modular scoring instrument derived from the gold answer, source-supported doctrine, expected failure modes, and, when available, observed response clusters.

Karthic first audits the packet fields and marks them as fixed, fixed but jurisdiction-sensitive, or needing confirmation. It then locks the doctrinal path by stating the controlling doctrine, gate order, benchmark posture, and strongest counterargument. Rubric rows are organized into modules for metadata, structural gatekeeping, primary doctrine gates, fallback doctrines and defenses, and cross-cutting answer discipline. Scored rows require observable legal behavior, not stylistic preferences.

Each full row card should contain a row code, module, weight, description, non-applicability guidance, golden target summary, observable features of a strong answer, allowed omissions, contradiction flags, comparison guidance, scoring anchors, failure labels, and row status. Broad rows are split only when the split improves legal discrimination. Overlapping rows require a distinctness rationale so that one legal error is not counted twice. Weights are assigned by legal importance rather than prose length, and answer-level penalties or caps are applied only after row scoring.

D. Dasha Context

Dasha’s instruction context defines reasoning-signature extraction and centroid-level evaluation. Dasha does not write the question, gold answer, or rubric. It reads raw model responses after generation and extracts a normalized description of the legal path each response appears to use. The current Dasha schema separates the controlling `primary_reasoning_path` from `secondary_paths`. Each secondary path records the gate or theory, its posture, the reason it mattered, and its effect on the outcome. This lets Dasha represent answers that discuss several possible gates before settling on one path.

The Dasha track audit records which question version is active, which rubric belongs to that question, whether the run is single-track or perturbation-track, which judge model is used, whether judge aggregation is single-model or ensemble-based, and whether the judge is separate from the response-model pool. Missing inputs that would distort scoring are review flags rather than fields to silently invent.

Dasha’s primary scored unit is the centroid. A centroid receives row-level scores under the Karthic rubric. If perturbation tracks are active, original-question outputs are clustered and scored only against the base track, and variation outputs are clustered and scored only against their corresponding variation track. Comparisons between original and variation tracks occur only after both tracks have been independently clustered.

Dasha also records centroid composition metadata: cluster size, per-model answer counts and shares, represented model count, dominant model, and dominant model share. This metadata is descriptive and must not become a scoring input. Dasha also records the full secondary-path audit profile for a centroid; the grouping key uses Dasha’s primary canonical fields so accepted, rejected, uncertain, and background theories remain visible without automatically fragmenting clusters.

E. Judge and Zak Context

The judge context requires row-level rubric application. For every cluster representative, the judge receives the representative response, the cluster legal signal, and the Karthic rubric rows. It returns exactly one score and rationale for every row. The score scale, weighting rule, and aggregation rule are fixed in the run configuration.

Zak is the escalation layer. It creates packets only when the frozen thresholds indicate uncertainty, disagreement, mixed clusters, missing rows, or other configured failure states. An escalation packet contains the active track, disputed clusters, relevant rubric rows, score summaries, response excerpts, reason for escalation, and a concise handoff note. If no escalation is required, the run still records an explicit no-escalation state.

Appendix B Artifact Examples

This appendix shows concrete artifacts from the live internal-validation run. The examples are generated from the run bundle rather than rewritten by hand.

A. Frank Packet Example

The following fields are direct excerpts from the live Frank packet. They show the source-derived scenario, expected answer, and the perturbation tracks that later generated model responses.

Frank field	Live output excerpt
Source	cases/MarriageSoF_Anglemire v Policemens Benev Assn of Chicago.pdf
Doctrine family	Statute of Frauds - agreements made upon consideration of marriage (oral promise to confer property/beneficiary status in exchange for marriage)

Neutral question	In Illinois, Clinton has long been a member of a fraternal benefit association that pays a death benefit to the member's named beneficiary. Under the association's constitution and bylaws, a member may change beneficiaries. The bylaws also provide that if the member cannot surrender the existing certificate for cancellation, he may instead file an affidavit explaining why he cannot produce it and naming a new beneficiary; upon that affidavit, the association will cancel the old certificate and issue a new one that is binding on the member, the beneficiaries, and the association. Before marrying Amanda, Clinton orally tells her: "If you marry me, I will surrender my current certificate and obtain a new certificate naming you as beneficiary." Amanda marries Clinton relying on that promise. After the marriage, Clinton has the association issue a new certificate naming Amanda as beneficiary, and Amanda keeps that certificate. About a year later, after serious marital conflict, Clinton asks the association to change beneficiaries again. He tells the association he cannot produce the certificate because Amanda has it, signs the associa...
Gold answer	Bottom-line outcome: No. The premarital oral promise made in consideration of marriage is void/unenforceable under the Statute of Frauds, and the subsequent marriage (and even the later voluntary act of initially naming Amanda as beneficiary) does not remove the promise from the statute. Therefore Clinton could later change beneficiaries in compliance with the association bylaws, and the later-named beneficiaries take. Controlling doctrine: Statute of Frauds provision requiring a writing for agreements made upon consideration of marriage (other than the mutual promise to marry). Writing requirement and trigger: Illinois requires that an agreement made upon consideration of marriage be in writing and signed by the party to be charged. The trigger is satisfied here because Clinton's promise was explicitly conditioned on Amanda marrying him (consideration of marriage) and concerned settl...
Variation V_surface_invariant (invariant)	Changed fact: Only party and organization names are changed; all legally operative facts, dates, writings, performances, and later disputes remain the same. Expected behavior: answer_invariant; the dominant legal reasoning and outcome should not change. Question: In Illinois, Jordan has long been a member of a fraternal benefit association that pays a death benefit to the member's named beneficiary. Under the association's constitution and bylaws, a member may change beneficiaries. The bylaws also provide that if the member cannot surrender the existing certificate for cancellation, he may instead file an affidavit explaining why he cannot produce it and naming a new beneficiary; upon that affidavit, the association will cancel the old certificate and issue a new one that is binding on the member, the beneficiaries, and the association. Before marrying Alex, Jordan orally tells her: "...
Variation V1 (material)	Changed fact: Before the wedding, Clinton signs a short written note stating: "In consideration of Amanda marrying me, I agree to name her beneficiary on my association certificate," and signs it. Expected behavior: Model should recognize that the marriage-consideration agreement is no longer barred for lack of writing (likely satisfying the SOF), but must still analyze whether that enforceable agreement actually restricts Clinton's contractual/bylaw power to change beneficiaries (and what remedy follows). The outcome is not fully resolved by the source; answer should be conditional and flag that the source chiefly addresses the no-writing scenario. Question: In Illinois, Clinton is a member of a fraternal benefit association whose bylaws allow him to change beneficiaries and also allow a change by affidavit if he cannot surrender the existing certificate. Before marrying Amanda, Clin...

Variation V2 (material) Changed fact: Clinton’s promise is not conditioned on marriage; he promises after engagement: ”Regardless of whether we marry, I’ll name you beneficiary,” and Amanda later marries for independent reasons. Expected behavior: Question: In Illinois, Clinton has long held a death-benefit certificate in a fraternal benefit association whose bylaws allow him to change beneficiaries (including by affidavit if he cannot surrender the existing certificate). While engaged to Amanda, Clinton orally tells her: ”Even if we never marry, I’m going to name you as beneficiary on my association certificate.” Amanda does not promise anything in return, and her later decision to marry Clinton is for independent personal reasons, not as consideration for the beneficiary promise. After the marriage, Clinton has the association issue a new certificate naming Amanda as beneficiary, and she keeps the certific...

B. Karthic Rubric Example

Karthic generated the following row-level scoring instrument from the locked Frank packet. Each row is later applied by Judge to Dasha cluster representatives.

Row	Category	Weight	Criterion	Source support excerpt
R0	doctrine	0.16	Correctly identifies the controlling doctrine as the Illinois Statute of Frauds provision for agreements made upon consideration of marriage (beyond mutual promises to marry) and frames the dispute as enforcement of a premarital bargain exchanging marriage for a property/benefit interest (beneficiary status / restriction on later change).	Frank gates G1; source_evidence: statute language quoted in packet (“no action... on any agreement made in consideration of marriage, unless... in writing and signed”; “other than the mutual promise to marry”); source_ext...
R1	rule	0.14	States the governing rule: a marriage-consideration agreement is unenforceable unless there is a writing/signed memorandum by the party to be charged (or authorized agent).	Gate G2 rule and source_evidence; gold_answer section “Writing requirement and trigger” referencing Ill. Rev. Stat. ch. 59 § 1.
R2	facts	0.1	Accurately uses the legally triggering facts: the promise was premarital, explicitly conditioned on marriage (“If you marry me...”), and aimed at obtaining/maintaining beneficiary status on the association certificate; acknowledges there is no signed premarital writing/memorandum.	source_extraction trigger_facts (premarital oral promise conditioned on marriage; no written agreement); neutral_question last paragraph (“Assuming there is no written premarital agreement...”).

R3	rule	0.14	Applies the writing requirement to conclude the premarital promise is barred/void/unenforceable as a parol marriage-consideration agreement (i.e., failure of SOF compliance).	Gate G2; gold_answer bottom line ("premarital oral promise... void/unenforceable under the Statute of Frauds"); source_evidence that court treated the promise as "parol/oral and therefore void."
R4	exceptions	0.14	Correctly rejects part performance/estoppel based on the marriage (and related reliance) as a basis to escape the marriage-consideration Statute of Frauds: marriage after the promise is not part performance for this category and is the precise scenario the statute contemplates.	Gate G3 rule and quoted opinion language in packet ("subsequent marriage... is not... part performance"; "Promise of marriage, followed by the marriage, is the exact situation contemplated by the statute"); gold_answer "C..."
R5	exceptions	0.1	Addresses the post-marriage certificate initially naming the spouse and explains (consistent with the source) that it is treated as a voluntary act that does not supply enforceable consideration-based rights that override the Statute of Frauds bar on the premarital bargain.	Gate G4 source_evidence ("any subsequent act... purely voluntary"); gold_answer explains initial designation does not remove statute bar.
R6	rule	0.1	Separately applies the association constitution/bylaws: if no enforceable contrary obligation exists, the member retains the right to change beneficiaries using the bylaw mechanism (including affidavit when unable to surrender the certificate), making the later-issued certificate effective as to the fund.	Gate G4 rule and source_evidence ("insured had the right to change the beneficiary... as provided by the constitution and by-laws"; bylaws allowed change by affidavit); source_extraction trigger_facts describing affidav...
R7	counterargument	0.06	Presents and fairly answers the strongest pro-spouse counterargument: that the arrangement became an "executed contract" after marriage and initial designation, or that equity should estop a later change due to reliance; then explains why the case's reasoning rejects that move for marriage-consideration agreements.	gold_answer "Strongest counterargument" and rejection rationale tied to G3/G4 (marriage not part performance; post-marriage act voluntary).

R8	conclusion	0.04	Gives a clear bottom-line allocation of proceeds consistent with the source-grounded outcome: spouse cannot enforce the premarital promise to defeat the later bylaw-compliant change; later-named beneficiaries take.	gold_answer bottom line; source_extraction holding summary (appellate court held promise void; member could change beneficiaries under bylaws).
R9	variation	0.02	Shows sensitivity to legally meaningful variations flagged by the packet (e.g., a signed premarital memorandum could satisfy the writing requirement; a promise not conditioned on marriage may not trigger the marriage-consideration SOF gate) and avoids overstating the source beyond the no-writing scenario.	variations V1 (signed note; outcome not fully resolved by source) and V2 (promise not conditioned on marriage); source_limits noting Illinois-specific and that the case chiefly addresses the no-writing scenario.

C. Model Response Examples

Model	Dasha cluster	Response excerpt
gpt-5.2	original__cluster	Nb. ### 1) The premarital promise is within the Illinois Statute of Frauds and is unenforceable without a writing In Illinois, an oral promise **made in consideration of marriage** ("If you marry me, I will ...") falls within the Statute of Frauds (750 ILCS 80/1, traditional "marriage consideration" provision). A promise to confer a property benefit on a spouse in exchange for marrying-such as a promise to keep the spouse as beneficiary of a death benefit-must be in a signed writing to be enforceable. Here, the promise was explicitly conditioned on marriage and there is **no signed premarital agreement or memorandum** , so the underlying contract claim fails. ### 2) Marriage (and related rel...
gpt-5.2	original__cluster	Nb. ### 1) The premarital promise is within the Statute of Frauds In Illinois, an oral promise made *in consideration of marriage* ("marry me and I will do X") falls within the Statute of Frauds and is unenforceable unless there is a writing signed by the promisor (or an equivalent signed memorandum). A promise to name (or keep) a spouse as beneficiary because she marries him is the classic "marriage consideration" contract. So, absent a signed writing, Amanda cannot sue to enforce Clinton's premarital promise as a contract. ### 2) Marriage (and related reliance) does not supply part performance to avoid that bar Illinois does not treat the marriage itself as "part performance" that takes...
gpt-5.4	original__cluster	Nb. Under Illinois law, Amanda would almost certainly lose, and the later bylaw-compliant change of beneficiary would control. ## Why ### 1. The premarital promise is within the Statute of Frauds A promise made **in consideration of marriage** -for example, "marry me and I'll give you this property/right/benefit"-must generally be **in writing** to be enforceable, unless it is merely a mutual promise to marry. Clinton's statement was not just "I'll marry you." It was a collateral promise: if Amanda married him, he would make her beneficiary of the fraternal benefit certificate. That is the kind of marriage-based promise Illinois treats as falling within the Statute of Frauds. Because there...

gpt-5.4	original__cluster Nb. Under Illinois law, Amanda would almost certainly lose, and the later bylaw-compliant change of beneficiary would control. ## Why ### 1. The premarital promise is within the Statute of Frauds A promise made **in consideration of marriage**-other than the mutual promise to marry itself-must be in writing to be enforceable. Clinton's statement was essentially: - **If you marry me, I will make you beneficiary.** That is a classic **promise upon consideration of marriage**. Since there is: - no written premarital agreement, and - no signed memorandum, Amanda cannot enforce that promise as a contract. ## 2. Marriage itself is not enough part performance Illinois follows the rule that **mar...
claude-sonnet-4-6	original__cluster #1 Amanda's Claim to the Death Benefit: Legal Analysis ## Overview of the Problem This question sits at the intersection of several doctrines: the Statute of Frauds (particularly the marriage provision), promissory estoppel, part performance, the contractual rights of fraternal benefit association members, and the enforceability of beneficiary designations. The short answer is that **Amanda faces serious obstacles and will likely fail**, though the analysis is genuinely complex and some arguments have real force. — ## I. The Threshold Problem: Statute of Frauds ### A. The Marriage Provision Illinois follows the traditional Statute of Frauds, which requires that promises **made in consider...
claude-sonnet-4-6	original__cluster #3 Amanda's Claim to the Death Benefit: Legal Analysis ## Overview of the Problem This question sits at the intersection of four distinct legal doctrines: (1) the Statute of Frauds as applied to premarital promises, (2) promissory estoppel and part performance, (3) the contractual/quasi-contractual enforceability of beneficiary designations in fraternal benefit certificates, and (4) the internal governance rules of fraternal benefit associations and their legal effect on third-party claims. The short answer is that Amanda faces serious obstacles but has plausible arguments, and the outcome turns substantially on which theory she advances and how Illinois courts weigh competing consideratio...
claude-haiku-4-5-20251001	original__cluster #7 Analysis of Amanda's Claims to the Death Benefit This is a challenging case involving the intersection of contract law, the Statute of Frauds, promissory estoppel, and the nature of beneficiary designations. Amanda faces significant obstacles, though she has some arguable claims. ## I. The Oral Promise and Statute of Frauds ### The Core Problem Amanda's most direct claim-breach of Clinton's oral promise to name her beneficiary-runs headlong into the **Statute of Frauds**. Under Illinois law (following the Restatement): - **Marriage itself is a consideration** for the promise, so the promise falls within the marriage provision of the Statute of Frauds (contracts made in consideration of...
claude-haiku-4-5-20251001	original__cluster #6 Analysis of Amanda's Claims This is a nuanced problem involving the intersection of contract law, promissory estoppel, part performance doctrine, and the enforceability of beneficiary designations. Amanda faces substantial obstacles. ## I. The Statute of Frauds Problem ### A. Express Contract Claim Amanda's strongest argument is that Clinton made an enforceable contract: "Marry me and I'll name you beneficiary." However, this faces the **statute of frauds**. **Illinois recognizes a statute of frauds for:** - Contracts in consideration of marriage (Ill. Comp. Stat. 815 ILCS 105/2) - Contracts that cannot be performed within one year **Critical issue:** Is a promise to name someone as a b...

google/gemini-3-flash	original__cluster	Under Illinois law, Amanda likely cannot enforce the oral premarital promise to invalidate the later beneficiary change. Her claim would likely fail due to the Statute of Frauds and the Illinois Uniform Premarital Agreement Act (IUPAA) , and her arguments for equitable relief (part performance or estoppel) are generally unsuccessful in this specific context. Here is the legal breakdown of why Amanda would likely lose: ### 1. The Writing Requirement (IUPAA and Statute of Frauds) In Illinois, any agreement made in consideration of marriage must be in writing. * IUPAA (750 ILCS 10/) : Since 1990, Illinois has followed the Uniform Premarital Agreement Act. Section 3 of the Act e...
google/gemini-3-flash	original__cluster	Under Illinois law, Amanda will likely fail in her attempt to enforce the oral premarital promise or to invalidate the subsequent beneficiary change. The resolution of this case depends on the intersection of the Illinois Statute of Frauds, the Illinois Uniform Premarital Agreement Act (IUPAA), and the law governing beneficiary designations in fraternal benefit associations. ### 1. The Statute of Frauds and the IUPAA The primary obstacle for Amanda is the requirement that agreements made in consideration of marriage must be in writing. * The IUPAA (750 ILCS 10/1 et seq.) : Illinois adopted the Uniform Premarital Agreement Act in 1990. Section 3 of the Act explicitly states: "A prem...
google/gemini-3-pro	original__cluster	Based on Illinois law, no , Amanda cannot enforce the premarital promise or use estoppel/part performance to invalidate the later beneficiary change. The death benefit will be awarded to Clinton's children and sister. Here is the legal analysis of why Amanda's claim will fail: ### 1. The Statute of Frauds Under the Illinois Statute of Frauds (740 ILCS 80/1), any agreement made upon consideration of marriage must be in writing and signed by the party to be charged. Clinton's promise to name Amanda as his beneficiary was made orally and strictly in consideration of her marrying him. Because there is no written premarital agreement or signed memorandum, the contract is unenforceable. ###...
google/gemini-3-pro	original__cluster	Based on Illinois law, no , Amanda cannot enforce the premarital promise or use estoppel/part performance to invalidate the later beneficiary change. The death benefit will be awarded to Clinton's children and sister. Here is the legal analysis of why Amanda's claim will fail: ### 1. The Statute of Frauds Under the Illinois Statute of Frauds (740 ILCS 80/1), any agreement made in consideration of marriage must be in writing and signed by the party to be charged. Because Clinton's promise ("If you marry me, I will...") was entirely oral, it is presumptively unenforceable. ### 2. Marriage is Not "Part Performance" Amanda cannot rely on the act of marriage itself to trigger the "part perfor...
anthropic/claude-4.5-sonnet	original__cluster	This is a complex problem involving contract law, the Statute of Frauds, and the interaction between private agreements and organizational bylaws. Let me work through the key issues: ## The Statute of Frauds Problem Clinton's premarital promise faces a significant obstacle: the Statute of Frauds requires contracts made in consideration of marriage to be in writing. This is one of the classic categories covered by the Statute of Frauds in Illinois and virtually all jurisdictions. Amanda's theories to overcome this: ### 1. Part Performance Exception - Amanda actually married Clinton in reliance on the promise - Clinton initially performed by designating her as beneficiary - Courts a...

anthropic/claude-4.5-sonnet	original__cluster	This is a complex question involving contract law, the statute of frauds, and the rights of fraternal benefit association members. Let me break down the key issues: ## The Premarital Promise and Statute of Frauds Problem for Amanda: Clinton's oral premarital promise likely falls within the statute of frauds as: - A promise made in consideration of marriage (Illinois requires such promises to be in writing) - Under 740 ILCS 80/1, contracts in consideration of marriage must be in writing Potential exception: The part performance doctrine might apply: - Amanda fully performed by marrying Clinton - Clinton partially performed by initially naming her beneficiary - However, part...
openai/gpt-4.1-mini	original__cluster	This question involves issues of contract enforceability, the statute of frauds, estoppel/part performance, and the effect of the association's bylaws on beneficiary designations. — ### Facts summary: - Clinton is a member of a fraternal benefit association that pays death benefits to named beneficiaries. - The bylaws allow members to change beneficiaries by surrendering the certificate or, if unable, by affidavit explaining why the certificate cannot be produced. - Clinton orally promises Amanda before marriage that if she marries him, he will surrender his certificate and get a new one naming her beneficiary. - Amanda marries Clinton relying on this promise. - After marriage, Clinton...
openai/gpt-4.1-mini	original__cluster	Let's analyze the situation step-by-step under Illinois law principles related to contract formation, estoppel, and beneficiary designations in fraternal benefit associations. — ### Facts Summary: - Clinton has a certificate naming a beneficiary under a fraternal benefit association. - The association's bylaws allow changing beneficiaries either by surrendering the old certificate or by affidavit if surrender is impossible. - Clinton orally promises Amanda before marriage: if she marries him, he will surrender the old certificate and get a new one naming her beneficiary. - Amanda marries Clinton relying on this promise. - After marriage, Clinton obtains a new certificate naming Amanda b...
deepseek-ai/deepseek-r1	original__cluster	Based on Illinois law regarding fraternal benefit societies, the Statute of Frauds, and promissory estoppel, Amanda is unlikely to succeed in enforcing the premarital promise or invalidating the later beneficiary change. The children/sister should prevail. Here's the analysis: 1. The Statute of Frauds Bars Enforcement of the Contract: * Clinton's promise ("If you marry me, I will... name you as beneficiary") constitutes a promise made "in consideration of marriage." Under the Illinois Statute of Frauds (740 ILCS 80/1), such promises must be in writing and signed by the party to be charged (Clinton) to be enforceable. * There is no written premarital agreement or signed memoran...
deepseek-ai/deepseek-r1	original__cluster	Based on Illinois law governing fraternal benefit associations and contract principles, Amanda likely CANNOT enforce the premarital promise or invalidate the later beneficiary change to recover the proceeds. Here's the breakdown: 1. The Premarital Promise as a Contract: * Statute of Frauds for Premarital Agreements (750 ILCS 10/1 et seq.): Illinois law requires premarital agreements to be "in writing" and signed by both parties to be enforceable. Clinton's oral promise ("If you marry me, I will... name you as beneficiary") is precisely the type of promise concerning property rights upon marriage that falls within this statute. * No Signed Writing: There is no written premar...

meta/meta-llama-3-70b-instruct	original__cluster	A5 fascinating question! In this scenario, Amanda is relying on the doctrine of promissory estoppel to enforce Clinton's premarital promise. To succeed, Amanda must demonstrate that: 1. Clinton made a clear and unambiguous promise to name her as the beneficiary. 2. Amanda reasonably relied on that promise by marrying Clinton. 3. Her reliance was detrimental, meaning she gave up something of value (in this case, her freedom to marry someone else or not marry at all). 4. Injustice can be avoided only by enforcing the promise. Here, Amanda has a strong case: * Clinton's oral promise was clear and specific. * Amanda reasonably relied on the promise by marrying Clinton, which is a significant c...
meta/meta-llama-3-70b-instruct	original__cluster	What an interesting case! In Illinois, the court would likely apply the doctrine of promissory estoppel to determine whether Amanda can enforce Clinton's premarital promise. Promissory estoppel is a legal doctrine that allows a party to recover on a promise even if it's not supported by consideration (something of value exchanged) if the following elements are met: 1. A promise was made. 2. The promisor (Clinton) should have reasonably expected the promisee (Amanda) to act in reliance on the promise. 3. The promisee did act in reliance on the promise. 4. Injustice can be avoided only by enforcing the promise. In this case: 1. Clinton made an oral promise to name Amanda as the beneficiary...
gpt-5.2	V_surface_invariant	No. ## 1 1 The premarital promise is within the Statute of Frauds and is not enforceable without a writing In Illinois, a promise "made in consideration of marriage" (i.e., "marry me and I'll do X") falls squarely within the Statute of Frauds and is unenforceable unless there is a written, signed agreement or memorandum . Jordan's statement-"If you marry me, I will surrender my current certificate and obtain a new certificate naming you as beneficiary"-is exactly that kind of promise. ### 2) Marriage (and related reliance) does not take this kind of agreement out of the Statute of Frauds in Illinois Alex cannot use part performance or equitable estoppel based on (a) the mar...
gpt-5.2	V_surface_invariant	No. ## 1 1 The premarital promise is within Illinois's Statute of Frauds and is unenforceable without a writing Illinois requires a signed writing for an agreement made upon consideration of marriage (i.e., "marry me and I will do X"), other than mutual promises to marry. Jordan's statement-"If you marry me, I will ... obtain a new certificate naming you as beneficiary"-is exactly that kind of bargain. With no written premarital agreement or signed memorandum , Alex cannot enforce the promise as a contract to keep her as beneficiary. ### 2) Marriage (and the later initial designation) does not take this kind of promise out of the Statute of Frauds Illinois does not treat marriage i...
gpt-5.4	V_surface_invariant	No. Under Illinois law, Alex is very unlikely to enforce Jordan's premarital oral promise or use estoppel/part performance to defeat the later beneficiary change that complied with the association's bylaws. ## Why ### 1. The oral premarital promise is within the Statute of Frauds A promise made "in consideration of marriage"-if you marry me, I will do X"-must generally be in writing to be enforceable. Jordan's promise was exactly that: *marry me and I'll name you beneficiary.* Because there is no written premarital agreement or signed memorandum, Alex cannot enforce that promise as a contract. ### 2. Marriage itself is not enough part performance Illinois follows the rule that ma...

gpt-5.4	V_surface_invariantNo. Under Illinois law, Alex likely cannot enforce Jordan's premarital oral promise, and the later beneficiary change made in compliance with the association's bylaws will control. ## Why ### 1. The oral promise is within the Statute of Frauds A promise made "in consideration of marriage"-for example, "if you marry me, I will give you property / make you beneficiary"-must be in writing to be enforceable. Jordan's statement was exactly that type of promise: "If you marry me, I will ... obtain a new certificate naming you as beneficiary." Because there is no written premarital agreement or signed memorandum, Alex cannot enforce that promise as a contract. ## 2. Marriage itself is no...
claude-sonnet-4-6	V_surface_invariant## Analysis of Alex's Claim Against the Death Benefit Proceeds ## Overview of the Legal Problem This question sits at the intersection of several distinct legal doctrines: the Statute of Frauds (particularly the marriage provision), equitable estoppel and part performance, the internal governance rules of fraternal benefit associations, and the competing equities between a surviving spouse and named beneficiaries. The answer is probably no-Alex almost certainly cannot prevail-though the analysis is genuinely complex and some equitable arguments deserve serious treatment before being rejected. — ## I. The Threshold Problem: The Statute of Frauds ### A. The Marriage Provision Illinois fo...
claude-sonnet-4-6	V_surface_invariant## Analysis of Alex's Claim Against the Death Benefit Proceeds ## Executive Summary Alex faces a genuinely difficult case with significant doctrinal obstacles, but has meaningful arguments worth carefully analyzing. The honest assessment is that Alex likely loses under orthodox Illinois law, but the claim is not frivolous and a sympathetic court could rule otherwise under equitable theories. Here is the full analysis. — ## I. The Threshold Problem: The Statute of Frauds ### A. The Promise Falls Within the Marriage Provision Jordan's promise - "marry me and I'll name you beneficiary" - is a promise made "in consideration of marriage". Under the Illinois Statute of Frauds (740 ILCS 80/1), agreem...
claude-haiku-4-5-20251001	V_surface_invariant## Analysis of Alex's Claim to the Death Benefit This is a challenging case involving the intersection of contract law, the Statute of Frauds, equitable estoppel, and the nature of beneficiary designations. Alex faces significant obstacles but has some arguable claims. ## I. The Oral Contract for a Beneficiary Designation ### A. Statute of Frauds Problem The threshold issue is whether the oral promise is barred by the Statute of Frauds. Illinois courts would likely analyze this under two potential provisions: 1. Marriage provision (Ill. Comp. Stat. 815 ILCS 105/3): Promises "in consideration of marriage" must be in writing. Jordan's promise-"If you marry me, I will name you as bene...
claude-haiku-4-5-20251001	V_surface_invariant## Analysis of Alex's Claim to the Death Benefit This is a genuinely difficult question that turns on several intersecting doctrines. The answer is likely no, Alex cannot recover, but with meaningful caveats about estoppel theory. ## I. The Core Problem: Statute of Frauds ### A. The Oral Promise and SOF Illinois recognizes that promises made in consideration of marriage are subject to the statute of frauds (Ill. Comp. Stat. 815 ILCS 105/2). The promise-"marry me and I'll name you beneficiary"-is classically within SOF because it's: - Made in consideration of marriage (the marriage itself is the consideration) - Not capable of full performance within one year (the designation is perpetu...

google/gemini-3-flash	V_surface_invariant	Under Illinois law, Alex will likely **not** be able to enforce the oral premarital promise to invalidate the later beneficiary change. The primary obstacles for Alex are the **Statute of Frauds** , the **Illinois Uniform Premarital Agreement Act (IUPAA)** , and the specific way Illinois courts treat marriage as "consideration." Here is the legal breakdown of why Alex's claim would likely fail: ### 1. The Statute of Frauds and the IUPAA In Illinois, any agreement made in consideration of marriage must be in writing to be enforceable. **The Statute of Frauds (740 ILCS 80/1):** This long-standing rule requires that any promise made "upon consideration of marriage" must be in writing and sig...
google/gemini-3-flash	V_surface_invariant	Under Illinois law, Alex will likely **not** be able to enforce the oral premarital promise or invalidate the later beneficiary change. The resolution of this issue depends on the intersection of the Statute of Frauds, the Illinois Uniform Premarital Agreement Act (IUPAA), and the law governing fraternal benefit associations. ### 1. The Statute of Frauds and the IUPAA In Illinois, agreements made in consideration of marriage are subject to strict writing requirements: **The Illinois Frauds Act (740 ILCS 80/1):** This statute provides that no action shall be brought to charge any person upon any agreement made upon consideration of marriage, unless the agreement is in writing and signed...
google/gemini-3-pro	V_surface_invariant	Based on Illinois law, **Yes, Alex can recover the proceeds.** Although the initial oral promise was subject to the Statute of Frauds, Jordan's subsequent act of actually naming Alex as the beneficiary and delivering the certificate to her constituted **full execution** of the contract. This removed the agreement from the Statute of Frauds and gave Alex a **vested equitable interest** in the death benefit that Jordan could not unilaterally defeat. Here is the step-by-step legal analysis of how Illinois courts resolve this exact scenario: ### 1. The Statute of Frauds and "Part Performance" Under the Illinois Statute of Frauds (740 ILCS 80/1), any agreement made in consideration of marriage...
google/gemini-3-pro	V_surface_invariant	Based on Illinois law, **Alex cannot enforce the premarital promise or use estoppel/part performance to invalidate the later beneficiary change.** The death benefit will legally go to Jordan's children and sister. Here is the step-by-step legal analysis of why Alex's claims will fail: ### 1. The Nature of the Beneficiary's Interest Under Illinois law governing fraternal benefit associations and life insurance, a named beneficiary generally has a "mere expectancy" rather than a vested right to the proceeds. The member retains the absolute right to change the beneficiary at any time, provided they comply with the association's constitution and bylaws. The only exception to this rule is if t...
anthropic/claude-4.5-sonnet	V_surface_invariant	This is an excellent multi-layered question involving contract law, the statute of frauds, and the intersection of promissory estoppel with association governance rules. ## Alex's Likely Outcome: **Cannot Recover** Despite some equitable arguments, Alex will likely lose because: ### 1. **The Premarital Promise Issue is Moot** Alex doesn't need to rely on the oral premarital promise because **Jordan actually performed it** -he did change the beneficiary designation to Alex after marriage. The promise was fulfilled, so statute of frauds issues (which would apply to premarital promises in many jurisdictions) become irrelevant. ### 2. **Alex Received a Revocable Designation** The critical prob...

anthropic/claude-4.5-sonnet	V_surface_invariant	This is a complex question involving the intersection of contract law, the statute of frauds, promissory estoppel, and the internal rules of a fraternal benefit association. Here's my analysis: ## The Core Legal Issues ### 1. **Statute of Frauds Problem** Jordan's premarital promise likely falls under the statute of frauds as: - A **promise made in consideration of marriage** (one of the classic categories) - Illinois, like most states, requires such promises to be in writing **Alex's problem**: The oral promise alone is likely unenforceable under the statute of frauds. ### 2. **Part Performance/Promissory Estoppel Arguments** Alex has stronger arguments here: **Part Performance Exception...
openai/gpt-4.1-mini	V_surface_invariant	This is a question involving contract enforceability, estoppel, and the effect of the fraternal benefit association's bylaws on beneficiary designations. — ### Facts Summary: - Jordan was a member of a fraternal benefit association that pays death benefits to designated beneficiaries. - The association's bylaws allow changing beneficiaries by surrendering the old certificate or, if the certificate cannot be surrendered, by an affidavit explaining why and naming a new beneficiary. - Jordan orally promised Alex before marriage that he would surrender the old certificate and name Alex as beneficiary if she married him. - Alex married Jordan relying on that promise. - After marriage, Jordan...
openai/gpt-4.1-mini	V_surface_invariant	This question involves Illinois contract and insurance law principles, focusing on: - The enforceability of an oral premarital promise concerning beneficiary designation, - Whether estoppel or part performance can validate that promise despite the lack of a writing, - The effect of the association's bylaws on beneficiary changes, - And the competing claims to the death benefit after the last beneficiary change. — ### Key Facts and Issues: - Jordan was a long-time member of a fraternal benefit association. - The association's constitution/bylaws allow a member to change beneficiaries via surrender of certificate or affidavit if surrender is impossible. - Jordan promised Alex orally before...
deepseek-ai/deepseek-r1	V_surface_invariant	Based on Illinois law, **Alex cannot enforce Jordan's premarital promise or use estoppel/part performance to invalidate the later beneficiary change.** The children/sister are likely entitled to the death benefit. Here's the analysis: 1. **Statute of Frauds Bars Enforcement of the Premarital Promise:** * Jordan's oral promise ("If you marry me, I will... name you as beneficiary") is a promise to make a testamentary disposition (a disposition taking effect at death) or a conditional gift in contemplation of marriage. * Illinois' Statute of Frauds (740 ILCS 80/1, derived from the English Statute of Frauds) **requires promises to make testamentary dispositions or gifts in consideration of ma...
deepseek-ai/deepseek-r1	V_surface_invariant	Based on Illinois law, **Alex cannot enforce the premarital promise or use estoppel/part performance to invalidate the later beneficiary change.** The children/sister are entitled to the death benefit. Here's the analysis: 1. **Unenforceability of the Premarital Promise:** * Jordan's oral promise ("marry me and I'll name you beneficiary") is a promise to make a future disposition of property (the death benefit) in consideration of marriage. * Under Illinois law (the **Statute of Frauds**), such agreements must be **in writing and signed by the party to be charged** (Jordan) to be enforceable. *See 740 ILCS 80/1; In re Estate of Sewart*, 274 Ill. App. 3d 298 (1995). * Since there's no writ...

meta/meta-llama-3-70b-instruct	V_surface_invariant_cluster_4	An interesting question! In this scenario, Alex is arguing that Jordan's premarital promise to name her as beneficiary should be enforced, despite the later change in beneficiaries. To succeed, Alex would need to rely on the doctrine of promissory estoppel or part performance. Promissory estoppel is a legal doctrine that allows a party to recover benefits conferred in reliance on a promise, even if there is no formal contract. To establish promissory estoppel, Alex would need to show that: 1. Jordan made a clear and unambiguous promise to name her as beneficiary. 2. Alex reasonably relied on that promise by marrying Jordan. 3. Alex suffered a detriment (i.e., she married Jordan) in reliance...
meta/meta-llama-3-70b-instruct	V_surface_invariant_cluster_5	What an interesting scenario! In this case, Alex is arguing that Jordan's premarital promise to name her as beneficiary should be enforced, despite the lack of a written agreement. Alex is relying on the doctrine of promissory estoppel, which allows a party to recover damages or enforce a promise when they have reasonably relied on that promise to their detriment. To succeed, Alex must show that: 1. Jordan made a clear and unambiguous promise to name her as beneficiary. 2. Alex reasonably relied on that promise by marrying Jordan. 3. Alex's reliance was to her detriment (i.e., she would not have married Jordan without the promise). 4. Jordan's later attempt to change beneficiaries is unconscionable...
gpt-5.2	V1__cluster_10	Yes. The signed premarital note is enough (as a Statute of Frauds matter) to make the *marriage-consideration contract* enforceable, and that gives Amanda a basis to attack the later beneficiary change-though her success still turns on beneficiary/insurance rules and the association's change-of-beneficiary provisions. Under Illinois Statute of Frauds principles (as commonly stated): - **A promise "in consideration of marriage" must be in a signed writing to be enforceable.** Clinton's dated, signed note-"In consideration of Amanda marrying me, I agree to name her beneficiary..."-fits that category. It identifies the parties, states the marriage consideration, and states the essential promise...
gpt-5.2	V1__cluster_9	Yes. Under Illinois Statute of Frauds principles, the signed premarital writing is what makes Amanda's "marriage-consideration" agreement enforceable, and that enforceability is what gives her a basis to attack the later beneficiary change. - **Agreement "in consideration of marriage" must be in writing and signed.** A promise made in consideration of marriage (other than mutual promises to marry) falls within the Statute of Frauds. Clinton's signed, dated note saying, in substance, "in consideration of Amanda marrying me, I agree to name her beneficiary" is the kind of memorandum that satisfies that requirement. So **Amanda can enforce the agreement**. - **If there were no signed writing...
gpt-5.4	V1__cluster_2	Yes. The signed premarital writing is the key fact that can take Amanda's marriage-consideration agreement out of the Statute of Frauds problem. ### Why An agreement made **in consideration of marriage** generally falls within the **Statute of Frauds** and must be supported by a **writing signed by the party to be charged**. Here, before the marriage, Clinton signed a note stating: > "In consideration of Amanda marrying me, I agree to name her beneficiary on my association certificate." That is strong evidence of a signed written memorandum of the agreement. So, unlike a purely oral antenuptial promise, this promise is likely **enforceable** under Statute of Frauds principles. ### Effect...

gpt-5.4	V1__cluster_3	Yes. The signed premarital writing is highly important, because it likely satisfies the Statute of Frauds requirement for an agreement made in consideration of marriage. ## Why A promise made **in consideration of marriage** generally falls within the Statute of Frauds and must be **in writing and signed by the party to be charged** . Here, before the marriage, Clinton signed and dated a writing stating: > "In consideration of Amanda marrying me, I agree to name her beneficiary on my association certificate." That is strong evidence of an enforceable marriage-consideration contract, because: - it is **written** , - it is **signed by Clinton** , - it identifies the **consideration** (Amanda m...
claude-sonnet-4-6	V1__cluster_8	# Analysis of the Premarital Writing Under Statute of Frauds Principles ## The Core Statute of Frauds Issue The critical question is whether Clinton's promise to name Amanda as beneficiary - made **in consideration of marriage** - falls within the Statute of Frauds, which in most jurisdictions (including Illinois) requires that agreements made **in consideration of marriage** be in writing to be enforceable. — ## The Writing Clinton Signed Clinton's note stated: > **"In consideration of Amanda marrying me, I agree to name her beneficiary on my association certificate."* This document must be evaluated against Statute of Frauds requirements for sufficiency of a writing. A memorandum satis...
claude-sonnet-4-6	V1__cluster_10	# Analysis of the Premarital Writing Under Statute of Frauds Principles ## The Core Issue The question is whether Clinton's signed premarital note - promising to name Amanda as beneficiary **in consideration of her marrying him** - satisfies Statute of Frauds requirements and thereby strengthens Amanda's ability to enforce the agreement and challenge the subsequent beneficiary change. — ## The Relevant Statute of Frauds Concern Under traditional Statute of Frauds principles, a **promise made in consideration of marriage** (other than a mutual promise to marry) must generally be **in writing and signed** by the party to be charged to be enforceable. This is the so-called **"marriage cons...
claude-haiku-4-5-20251001	V1__cluster_7	# Analysis Under Illinois Statute of Frauds This question involves the intersection of the Statute of Frauds, marriage consideration, and beneficiary designation law. Here's the analysis: ## The Statute of Frauds Issue Under Illinois law, an agreement that is made "in consideration of marriage" must be in writing to be enforceable-this is a classic Statute of Frauds category. **Clinton's signed, dated note satisfies this requirement.** The writing: - Is signed by Clinton (the party to be charged) - Is dated - Clearly expresses the agreement - Explicitly references marriage consideration ("In consideration of Amanda marrying me") ## The Marriage Consideration Doctrine Illinois recognizes t...
claude-haiku-4-5-20251001	V1__cluster_6	# Analysis Under Illinois Statute of Frauds This question requires examining whether the premarital writing satisfies the Statute of Frauds and whether it affects Amanda's enforceability claims. ## The Statute of Frauds Issue Under Illinois law, agreements made in consideration of marriage must satisfy the Statute of Frauds and be in writing. The signed premarital note stating Clinton's agreement to name Amanda as beneficiary appears to satisfy this requirement because: 1. **Writing exists** : Clinton signed and dated a written note 2. **Clear terms** : It explicitly states the agreement and identifies the consideration (Amanda's marriage) 3. **Signature** : Clinton's signature is present ##...

google/gemini-3-flash	V1__cluster_4	Under Illinois law and general principles of equity regarding fraternal benefit associations, the signed premarital writing **significantly affects** the outcome by satisfying the Statute of Frauds and establishing that Amanda has an **equitable vested interest** in the death benefit. The analysis breaks down as follows: ### 1. Satisfaction of the Statute of Frauds In Illinois, the Statute of Frauds (740 ILCS 80/1) provides that any agreement made upon consideration of marriage must be in writing and signed by the party to be charged. * **The Writing:** Clinton signed and dated a note stating, "In consideration of Amanda marrying me, I agree to name her beneficiary on my association certi...
google/gemini-3-flash	V1__cluster_4	Under Illinois law and general principles of insurance equity, the signed premarital writing **does affect the outcome by satisfying the Statute of Frauds, thereby allowing Amanda to enforce the agreement and successfully challenge the later beneficiary change.** Here is the legal breakdown of why the writing is decisive in this scenario: ### 1. Satisfaction of the Statute of Frauds In Illinois, the Statute of Frauds (740 ILCS 80/1) requires that any agreement made "upon consideration of marriage" must be in writing and signed by the party to be charged. * **The Writing:** Clinton's note specifically states the consideration ("In consideration of Amanda marrying me") and the promise ("I a...
google/gemini-3-pro	V1__cluster_4	**Yes, the signed premarital writing fundamentally affects the outcome of this dispute.** Because of the signed writing, Amanda will be able to enforce the agreement, successfully challenge the later beneficiary change, and claim the death benefit. Here is how the Statute of Frauds and Illinois equitable principles apply to these facts: ### 1. Satisfaction of the Statute of Frauds Under the Illinois Statute of Frauds (740 ILCS 80/1), any agreement made "in consideration of marriage" must be in writing and signed by the party to be charged to be legally enforceable. * **The Problem without a Writing:** If Clinton had only made an oral promise to name Amanda as his beneficiary in exchange f...
google/gemini-3-pro	V1__cluster_4	**Yes.** The signed premarital writing fundamentally affects Amanda's rights. Because the writing satisfies the Statute of Frauds, Amanda can successfully enforce the agreement, establish a vested equitable interest in the certificate, and defeat the claims of the later-named beneficiaries. Here is a breakdown of how the legal principles apply to this specific scenario under Illinois law: ### 1. Satisfaction of the Statute of Frauds Under the Illinois Statute of Frauds, any agreement made in consideration of marriage must be in writing and signed by the party to be charged to be legally enforceable. * **Without the writing:** If Clinton had only made an oral promise to name Amanda as bene...
anthropic/claude-4.5-sonnet	V1__cluster_8	# Analysis of the Premarital Writing Under Statute of Frauds ## Does the Writing Satisfy the Statute of Frauds? **Yes, the signed premarital writing likely satisfies the Statute of Frauds** for the marriage-consideration promise, which affects Amanda's ability to enforce the agreement. ## Key Elements Present ### 1. **Writing Requirement - SATISFIED** - Clinton signed and dated a written note - The writing memorializes the essential terms of the agreement ### 2. **Marriage Consideration Exception - APPLICABLE** - The promise is "in consideration of marriage" - This falls within the traditional Statute of Frauds categories requiring a writing - Illinois follows this requirement ### 3. **Es...

anthropic/claude-4.5-sonnet	V1__cluster_2	# Analysis of the Premarital Writing Under Statute of Frauds ## The Core Issue The question asks whether Clinton's signed premarital note satisfies the Statute of Frauds for a marriage-consideration contract, potentially giving Amanda grounds to challenge the later beneficiary change. ## Statute of Frauds Requirements Under the Statute of Frauds, **contracts in consideration of marriage** must be in writing. The key requirements are: 1. A writing signed by the party to be charged (Clinton) 2. Containing essential terms 3. Showing consideration of marriage ## Analysis of Clinton's Note **What the writing provides:** - satisfied Signed and dated by Clinton (the promisor) - satisfied Explici...
openai/gpt-4.1-mini	V1__cluster_10	Let's analyze the situation step-by-step, focusing on Illinois law regarding beneficiary designations, contract enforceability, and the Statute of Frauds. — ### Facts Summary: - Clinton is a member of a fraternal benefit association. - The association's bylaws allow Clinton to change beneficiaries, including by affidavit if the original certificate cannot be surrendered. - Before marrying Amanda, Clinton signs a short written note: **"In consideration of Amanda marrying me, I agree to name her beneficiary on my association certificate."* - Amanda marries Clinton. - After marriage, Clinton obtains a certificate naming Amanda as beneficiary, which Amanda keeps. - One year later, Clinton cl...
openai/gpt-4.1-mini	V1__cluster_1	Let's analyze the facts and the legal issues step-by-step: — ### FACTS: - Clinton is a member of a fraternal benefit association. - The association's bylaws allow Clinton to change beneficiaries by: 1. Direct surrender of the certificate and re-issuance, or 2. By affidavit if he cannot produce the certificate. - Before marrying Amanda, Clinton signs and dates a short written note stating: **"In consideration of Amanda marrying me, I agree to name her beneficiary on my association certificate."* - Amanda marries Clinton. - After marriage, the association issues a certificate naming Amanda as beneficiary, and Amanda keeps it. - A year later, Clinton claims he cannot produce the certificate...
deepseek-ai/deepseek-r1	V1__cluster_5	The signed premarital writing **does affect** Amanda's ability to enforce the marriage-consideration agreement and challenge the later beneficiary change under Statute of Frauds principles. Here's why: 1. **Satisfaction of Statute of Frauds:** The premarital writing signed by Clinton ("In consideration of Amanda marrying me, I agree to name her beneficiary on my association certificate") **satisfies the Statute of Frauds requirement** for an agreement made in consideration of marriage. * It is **in writing**. * It clearly **identifies the subject matter** (naming Amanda beneficiary of the association certificate). * It **states the consideration** (Amanda marrying Clinton). * It is **sign...
deepseek-ai/deepseek-r1	V1__cluster_2	Based on the Statute of Frauds principles described: 1. **The Premarital Writing Satisfies the Statute of Frauds:** The signed, dated note stating "In consideration of Amanda marrying me, I agree to name her beneficiary on my association certificate" constitutes a sufficient writing to satisfy the Statute of Frauds for a contract made in consideration of marriage. It is in writing, signed by the party to be charged (Clinton), and identifies the consideration (marriage) and the subject matter (naming Amanda beneficiary). 2. **The Agreement is Enforceable:** Therefore, the agreement between Clinton and Amanda is enforceable. Amanda performed her obligation by marrying Clinton. Clinton parti...

meta/meta-llama-3-70b-instruct	V1__cluster_10	A fascinating question! Under the Statute of Frauds principles, a contract or agreement that falls within the statute’s scope must be in writing and signed by the party to be charged (in this case, Clinton). The premarital writing signed by Clinton, which states "In consideration of Amanda marrying me, I agree to name her beneficiary on my association certificate," can be considered a written agreement that satisfies the Statute of Frauds. This writing constitutes a promise to perform a specific act (naming Amanda as beneficiary) in exchange for Amanda’s performance of marrying Clinton. The writing is signed and dated, which meets the formal requirements of the Statute of Frauds. As a res...
meta/meta-llama-3-70b-instruct	V1__cluster_7	A classic contract law problem! Under the Statute of Frauds, a contract made in consideration of marriage must be in writing and signed by the party to be charged (in this case, Clinton) to be enforceable. The premarital writing signed by Clinton, "In consideration of Amanda marrying me, I agree to name her beneficiary on my association certificate," satisfies the Statute of Frauds requirements. This writing is a contractual agreement made in consideration of marriage, and it is in writing, signed, and dated. As a result, Amanda can enforce this agreement and challenge the later beneficiary change. The fact that Clinton later signed an affidavit and changed the beneficiaries does not nega...

D. Dasha Cluster Examples

Cluster	Member models	Outcome	Reasoning signature excerpt
original__cluster_1	anthropic/claude-4.5-sonnet (n=2), claude-sonnet-4-6, deepseek-ai/deepseek-r1 (n=2), google/gemini-3-flash, gpt-5.2, gpt-5.4 (n=2), openai/gpt-4.1-mini (n=2)	The oral premarital promise is void and unenforceable under the Statute of Frauds; the later bylaw-compliant beneficiary change is valid and the final-named beneficiaries take the proceeds	G1 satisfied (oral promise in consideration of marriage) \{\}(\{}rightarrow\{\}) G2 failed (no signed writing) \{\}(\{}rightarrow\{\}) G3 rejected (marriage is not part performance; estoppel unavailable) \{\}(\{}rightarrow\{\}) G4 applied (promise void; subsequent acts voluntary; bylaw-compliant change controls)
original__cluster_2	google/gemini-3-flash, google/gemini-3-pro (n=2), gpt-5.2	No. The oral premarital promise is void and unenforceable under the Statute of Frauds; Clinton’s later bylaw-compliant beneficiary change is effective; the final-certificate beneficiaries take the death benefit	Illinois Statute of Frauds requires a signed writing for agreements made upon consideration of marriage other than the mutual promise to marry; Clinton’s oral promise to name Amanda beneficiary was made in consideration of her marrying him; no signed writing exists; the promise is therefore void and cannot be enforced to bar a later beneficiary change
original__cluster_3	claude-sonnet-4-6	Amanda’s claim based on the oral premarital promise is unenforceable under the Statute of Frauds; the subsequent marriage does not constitute part performance sufficient to remove the promise from the statute; Clinton...	The oral premarital promise falls within the Illinois Statute of Frauds marriage-consideration clause; no writing exists; the subsequent marriage is the exact scenario the statute targets and does not constitute part performance; the post-marriage beneficiary designation was revocable under association bylaws; the oral promise is a nullity and Clinton retained the right to change beneficiaries

original__clustermeta/meta-llama-3-70b-instruct	Response concludes Amanda likely prevails under promissory estoppel and/or part performance, finding the oral promise enforceable and the later beneficiary change invalid	Response applies promissory estoppel as the controlling theory: Clinton made a promise, reasonably expected Amanda to rely on it, Amanda relied by marrying him, and injustice results without enforcement; response treats marriage as both reliance and consideration supporting the claim
original__clustermeta/meta-llama-3-70b-instruct	Amanda can enforce the oral premarital promise under promissory estoppel; the later beneficiary change is invalid and Amanda takes the death benefit	The response bypasses the Statute of Frauds writing requirement entirely by invoking promissory estoppel as a free-standing enforcement theory, treating marriage as sufficient detrimental reliance to satisfy all four estoppel elements and render the oral promise enforceable
original__clusterclaude-haiku-4-5-20251001	Express contract claim is barred by the Statute of Frauds for lack of a writing; promissory estoppel is treated as a surviving exception on liability, but practical recovery is limited to reliance damages rather than...	The response accepts that the oral promise falls within the Statute of Frauds marriage-consideration provision and that no writing exists, but treats promissory estoppel as a recognized Illinois exception that survives the statute on liability; however, the response concludes that even successful estoppel yields only reliance damages, not the death benefit, because the association lawfully followed its bylaws and the beneficiary designation was revocable absent an enforceable contractual restriction
original__clusterclaude-haiku-4-5-20251001	Amanda loses on breach of contract; the oral promise is unenforceable under the Statute of Frauds. Promissory estoppel is treated as a live but uncertain exception. Part performance is rejected as unlikely. Clinton's...	The oral promise falls within the Statute of Frauds marriage-consideration provision because it was conditioned on Amanda marrying Clinton and concerned a property-like benefit. No writing exists. The promise is therefore void as a contract. The subsequent marriage does not constitute part performance sufficient to remove the promise from the statute. Clinton retained the right to change beneficiaries under the association bylaws, and the later designation change is valid.
V_surface_invariantanthropic/claude-4.5-sonnet	Alex cannot recover; the later-named beneficiaries take because Jordan validly changed the designation under the association's bylaws	The response rests primarily on the revocability of beneficiary designations under the association's constitution and bylaws: because the bylaws expressly permit changes and Jordan followed the affidavit procedure, the later designation is valid and Alex holds only an expectancy, not a vested right

V_surface_invariant	anthropic/claude-4.5-sonnet, claude-haiku-4-5-20251001 (n=2), claude-sonnet-4-6 (n=2), deepseek-ai/deepseek-r1 (n=2), google/gemini-3-pro, openai/gpt-4.1-mini (n=2)	The oral premarital promise is unenforceable under the Statute of Frauds; Jordan retained the right to change beneficiaries under the association bylaws; Alex almost certainly cannot prevail	G1 (agreement in consideration of marriage) $\rightarrow$ G2 (no writing or signed memorandum satisfying the statute) $\rightarrow$ G3 (subsequent marriage is not part performance sufficient to remove the promise from the statute; it is the exact scenario the statute targets) $\rightarrow$ G4 (post-marriage beneficiary designation was voluntary and revocable; Jordan retained bylaw right to change beneficiaries; later-named beneficiaries take)
V_surface_invariant	google/gemma-3-flash (n=2), gpt-5.2 (n=2), gpt-5.4 (n=2)	The oral premarital promise is void and unenforceable under the Statute of Frauds; the later bylaw-compliant beneficiary change is valid and the subsequently named beneficiaries take	The promise is a marriage-consideration agreement under Illinois Statute of Frauds; no signed writing exists; the statute renders the oral promise a nullity; subsequent marriage does not constitute part performance for this category; all post-marriage acts are voluntary; the association bylaws independently govern beneficiary changes; the bylaw-compliant affidavit procedure was followed; the later certificate controls
V_surface_invariant	meta/llama-4-3-70b-instruct	Response concludes, or leaves open, that Alex may enforce the premarital promise via promissory estoppel or part performance, without resolving that the oral promise is void under the Statute of Frauds	Response frames the controlling question as whether promissory estoppel or part performance can enforce the oral promise, treating marriage as detrimental reliance and the initial beneficiary designation as partial performance by Jordan, without applying the Statute of Frauds bar or the rule that subsequent marriage does not constitute part performance for marriage-consideration agreements
V_surface_invariant	meta/llama-5-3-70b-instruct	Response does not apply the Statute of Frauds writing requirement or reach a definitive outcome; instead it frames the dispute as a promissory estoppel claim and leaves the result open-ended, contingent on court discr...	Response bypasses Statute of Frauds analysis entirely and anchors the claim in promissory estoppel, treating marriage as detrimental reliance on an oral promise rather than as the consideration that triggers the writing requirement

V_surface_invariant/google/gemini-36pro

Beneficiary (Alex) recovers; response concludes the oral contract was fully executed by actual delivery of the certificate, removing it from the Statute of Frauds and vesting an equitable interest that defeats the lat...

The response accepts that the oral promise is within the Statute of Frauds and that marriage alone is not part performance. It then pivots to an 'executed contract' exception: because Jordan actually named Alex and delivered the certificate, the contract was fully performed on both sides, removing the SOF bar. Full execution vested an equitable interest in Alex that Jordan could not unilaterally defeat, and the later bylaw-compliant change was ineffective against a prior vested equitable right; the later designees are mere volunteers subordinate to Alex as a purchaser for value.

V1__cluster_1 openai/gpt-4.1-mini

Amanda prevails; the signed premarital writing is enforceable under the SOF, and the later beneficiary change by affidavit is procedurally invalid because Amanda holds the original certificate, meaning the affidavit p...

The response finds that the signed premarital note satisfies the SOF writing requirement, making the marriage-consideration agreement enforceable; combined with Amanda's possession of the certificate and the invalidity of the affidavit procedure, Amanda's beneficiary status is protected

V1__cluster_2 anthropic/claude-4.5-sonnet, deepseek-ai/deepseek-r1, gpt-5.4

The marriage-consideration agreement is enforceable because the signed writing satisfies the Statute of Frauds; Amanda can challenge the later beneficiary change as a breach of that enforceable contract

G1 triggered (promise in consideration of marriage beyond mutual promise to marry) $\{\}(\{\}\rightarrow\{\})$ G2 satisfied (signed written memorandum exists) $\{\}(\{\}\rightarrow\{\})$ agreement enforceable $\{\}(\{\}\rightarrow\{\})$ later beneficiary change is breach of enforceable contract $\{\}(\{\}\rightarrow\{\})$ Amanda's claim survives

V1__cluster_3 gpt-5.4

Yes - the signed premarital writing removes the Statute of Frauds bar that would defeat a purely oral promise, giving Amanda a viable contractual and equitable basis to challenge the later beneficiary change, though t...

The response identifies that the source case turned on the absence of a writing (G2 not satisfied), then applies the same doctrine to the variation fact - a signed writing exists - and concludes G2 is now satisfied, removing the primary enforceability obstacle and allowing Amanda to assert a contractual obligation against Clinton and a potential constructive trust against later beneficiaries

V1__cluster_4	google/gemini-3-flash (n=2), google/gemini-3- pro (n=2)	Amanda prevails; the signed writing satisfies the Statute of Frauds, the marriage-consideration promise is enforceable, Amanda acquires a vested equitable interest that defeats Clinton's later bylaw-compliant benefici...	Signed writing by Clinton satisfies Illinois SOF writing-and-signature requirement for marriage-consideration agreements $\{\}(\{\}\rightarrow\{\})$ promise is enforceable $\{\}(\{\}\rightarrow\{\})$ marriage as valuable consideration vests equitable interest in Amanda $\{\}(\{\}\rightarrow\{\})$ Clinton's contractual surrender of change-right defeats later bylaw-compliant designation $\{\}(\{\}\rightarrow\{\})$ later beneficiaries are volunteers with no superior equitable claim $\{\}(\{\}\rightarrow\{\})$ Amanda entitled to death benefit
V1__cluster_5	deepseek-ai/deepseek-r1	The signed premarital writing satisfies the Statute of Frauds; the marriage-consideration agreement is enforceable; the later beneficiary change constitutes breach of that enforceable contract; Amanda has legal and eq...	Signed writing satisfies SOF writing requirement (G2) $\{\}(\{\}\rightarrow\{\})$ marriage-consideration agreement is enforceable $\{\}(\{\}\rightarrow\{\})$ Amanda performed by marrying $\{\}(\{\}\rightarrow\{\})$ Clinton's later beneficiary change via affidavit breaches the enforceable contract $\{\}(\{\}\rightarrow\{\})$ later beneficiaries take subject to Amanda's prior contractual and equitable claim
V1__cluster_6	claude-haiku-4-5- 20251001	Amanda's position is substantially strengthened; the signed writing likely satisfies the Statute of Frauds, the agreement is enforceable, and the later beneficiary change may constitute breach of contract	The response identifies a signed premarital note as satisfying G2 (writing and signature requirement), concludes the marriage-consideration agreement is enforceable, and reasons that the later beneficiary change breaches that enforceable contract
V1__cluster_7	claude-haiku-4-5- 20251001, meta/meta- llama-3-70b-instruct	Amanda prevails: the signed premarital writing is enforceable, creates a binding contractual obligation, and substantially challenges the later beneficiary change	The response pivots on G2 (writing requirement): because a signed, dated note explicitly referencing marriage consideration exists, the Statute of Frauds is satisfied and the oral-promise-void rule never applies. The response does not reach G3 (part performance) or G4 (voluntary-act effect) because it treats the writing as dispositive.
V1__cluster_8	anthropic/claude-4.5- sonnet, claude-sonnet-4-6	The signed premarital note satisfies the Statute of Frauds; Amanda is a contract beneficiary with enforceable rights that Clinton could not unilaterally defeat through a subsequent bylaw-compliant beneficiary change;...	The signed note identifies the parties, states the subject matter and essential terms, and is signed by the party to be charged, thereby satisfying the Statute of Frauds writing requirement; satisfaction of the writing requirement transforms Amanda from a revocable gratuitous designee into a contract beneficiary with vested rights Clinton could not unilaterally extinguish

V1__cluster_9	gpt-5.2	Where a signed writing exists satisfying the Statute of Frauds, the marriage-consideration agreement is enforceable and Amanda can challenge the later beneficiary change on contract and equitable grounds	Signed premarital note constitutes a memorandum satisfying the Illinois Statute of Frauds writing requirement for marriage-consideration agreements, making the agreement enforceable and removing the SOF barrier to Amanda's claim
V1__cluster_10	claude-sonnet-4-6, gpt-5.2, meta/meta-llama-3-70b-instruct, openai/gpt-4.1-mini	SOF satisfied by signed premarital note; marriage-consideration agreement enforceable; Amanda can challenge later beneficiary change on contract and equitable grounds	Signed dated note identifies parties, states marriage consideration, and states essential promise $\{\}(\{\}\rightarrow\{\})$ satisfies G2 writing requirement $\{\}(\{\}\rightarrow\{\})$ marriage-consideration agreement enforceable $\{\}(\{\}\rightarrow\{\})$ Clinton contractually bound not to change beneficiary $\{\}(\{\}\rightarrow\{\})$ later change is breach $\{\}(\{\}\rightarrow\{\})$ equitable remedy (constructive trust) available against later-named beneficiaries

E. Judge Scoring Examples

Judge receives a Dasha representative response, the cluster legal signal, and the Karthic rows. These rows show examples of the row-level scores and rationales that are projected to cluster members.

Cluster	Row	Score	Judge rationale excerpt
original__cluster_1R0		4.00	The response correctly identifies the controlling doctrine as the Illinois Statute of Frauds provision for agreements made in consideration of marriage, distinguishes it from a mere mutual promise to marry, and frames the dispute as enforcement of a premarital bargain exchanging marriage for beneficiary status on a fraternal benefit certificate. All key elements of R0 are present and accurately stated.
original__cluster_2R0		4.00	The response correctly identifies the Illinois Statute of Frauds as the controlling doctrine, frames the dispute as enforcement of a premarital bargain exchanging marriage for beneficiary status, and explicitly notes the promise was made 'in consideration of her marrying him.'
original__cluster_3R0		4.00	The framing matches the required doctrine precisely. The response correctly and explicitly identifies the Illinois Statute of Frauds (740 ILCS 80/1) as the controlling doctrine, frames the dispute as a premarital bargain exchanging marriage for beneficiary status, and distinguishes the marriage-consideration clause from the mutual promise to marry. The framing is precise and matches the rubric criterion fully.
original__cluster_4R0		0.00	The response never identifies the Illinois Statute of Frauds or its marriage-consideration provision. It frames the entire dispute as a promissory estoppel problem and does not mention the SOF at all, let alone the specific provision covering agreements made in consideration of marriage beyond mutual promises to marry.

original__cluster_5R0	0.00	The response never identifies the Illinois Statute of Frauds provision for marriage-consideration agreements. It frames the dispute entirely as a promissory estoppel matter and does not characterize the issue as enforcement of a premarital bargain triggering the SOF marriage-consideration clause.
original__cluster_6R0	2.50	The response identifies the statute of frauds as the controlling doctrine and frames the dispute as enforcement of a premarital bargain exchanging marriage for beneficiary status. However, it cites '815 ILCS 105/2' rather than the correct Illinois provision (Ill. Rev. Stat. ch. 59 § 1), does not quote or closely paraphrase the statutory language distinguishing 'other than the mutual promise to marry,' and does not sharply frame the doctrine as the marriage-consideration SOF category distinct from other SOF provisions. The framing is partially correct but imprecise.
original__cluster_7R0	3.00	The response correctly identifies the Illinois Statute of Frauds as the controlling doctrine and frames the dispute as enforcement of a premarital bargain exchanging marriage for beneficiary status. It notes that marriage is the consideration for the promise and that the promise falls within the marriage provision of the SOF. However, it does not explicitly quote or reference the statutory language distinguishing 'other than the mutual promise to marry,' and the framing is somewhat general rather than precisely tracking the Illinois statutory provision as described in the rubric.
V_surface_invariantR0_cluster_0.00	0.00	The response explicitly declares the premarital promise issue 'moot' and never identifies the Illinois Statute of Frauds provision for agreements made upon consideration of marriage as the controlling doctrine. The dispute is framed entirely around bylaw revocability, not the marriage-consideration SOF gate. Score 0 is better supported than score 2 because the criterion requires correctly identifying and framing the controlling doctrine, which the response affirmatively avoids.

F. Zak Escalation Example

Zak records when the judge layer should not be treated as silently final. The live run produced the following escalation packet.

Zak field	Live output excerpt
Packet id	zak_1
Question	Review disputed representative clusters.
Disagreement summary	Judge confidence did not meet the configured threshold.
Cluster ids	original__cluster_1, original__cluster_2, original__cluster_3, original__cluster_4, original__cluster_5, original__cluster_6, original__cluster_7, V_surface_invariant__cluster_1, V_surface_invariant__cluster_2, V_surface_invariant__cluster_3, V_surface_invariant__cluster_4, V_surface_invariant__cluster_5, V_surface_invariant__cluster_6, V1__cluster_1, V1__cluster_2, V1__cluster_3, V1__cluster_4, V1__cluster_5, V1__cluster_6, V1__cluster_7, V1__cluster_8, V1__cluster_9, V1__cluster_10
Rubric rows	R0, R1, R2, R3, R4, R5, R6, R7, R8, R9

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